Management Science

Publication details, including instructions for authors and subscription information:
http://pubsonline.informs.org

Equity Market Fragmentation and Capital Investment Efficiency

Wayne Landsman, Jing Pan, Stephen Stubben

To cite this article:
Wayne Landsman, Jing Pan, Stephen Stubben (2024) Equity Market Fragmentation and Capital Investment Efficiency. Management Science 70(7):4381-4406. https://doi.org/10.1287/mnsc.2023.4905

informs
https://pubsonline.informs.org/journal/mnsc
MANAGEMENT SCIENCE
Vol. 70, No. 7, July 2024, pp. 4381–4406
ISSN 0025-1909 (print), ISSN 1526-5501 (online)

Equity Market Fragmentation and Capital Investment Efficiency

Wayne Landsman,[a] Jing Pan,[b] Stephen Stubben[c,*]

[a] Kenan-Flagler Business School, University of North Carolina at Chapel Hill, Chapel Hill, North Carolina 27514; [b] Accounting Department, Smeal College of Business, Pennsylvania State University, University Park, Pennsylvania 16802; [c] University of Utah, Salt Lake City, Utah 84112

*Corresponding author

Contact: wayne_landsman@kenan-flagler.unc.edu (WL); jzp6150@psu.edu, https://orcid.org/0000-0002-1360-1636 (JP); stephen.stubben@eccles.utah.edu, https://orcid.org/0000-0001-6912-1490 (SS)

Received: September 19, 2021
Revised: May 5, 2022; November 9, 2022; December 18, 2022
Accepted: December 26, 2022
Published Online in Articles in Advance: September 1, 2023

https://doi.org/10.1287/mnsc.2023.4905

Abstract. This study examines how equity market fragmentation affects firms' capital investment decisions. Recent empirical research finds that market fragmentation lowers trading costs and thus improves market quality. We examine whether this increase in market quality translates into greater revelatory price efficiency, where stock prices reveal with greater precision information to managers and/or creditors about firms' investment opportunities. Consistent with this notion, our findings reveal that the association between capital investment and investment opportunities is increasing in market fragmentation. Additional evidence suggests that (a) market fragmentation increases revelatory price efficiency at least in part by encouraging information acquisition and informed trade by equity investors and (b) the more efficient stock prices inform both managers and creditors about firms' investment opportunities. Inferences based on difference-in-differences and instrumental variable tests are consistent with those based on our primary findings.

History: Accepted by Suraj Srinivasan, accounting.
Funding: The authors are grateful to the Kenan-Flagler Business School, Cox School of Business, Smeal School of Business, and David Eccles School of Business for funding our research.
Supplemental Material: The data files and online appendix are available at https://doi.org/10.1287/mnsc.2023.4905.

Keywords: equity market fragmentation • capital investment

1. Introduction

This study examines how equity market fragmentation affects firms' capital investment decisions. Market fragmentation is the extent to which trades are dispersed across multiple trading venues rather than concentrated in one or a small number of venues. In recent years, as the number of trading venues has increased, trading in U.S. equity securities has become more highly fragmented.[1] This change has led to concerns about the potential negative effects of market fragmentation on market quality. However, recent empirical research finds that market fragmentation improves market quality, as reflected by, for example, lower bid-ask spreads. We examine whether this increase in market quality translates into greater revelatory price efficiency, which is the extent to which stock prices reveal information useful for capital investment decisions.

Although the theoretical and empirical literature on market fragmentation has examined the effects of market fragmentation on dimensions of market quality, it has not investigated the effects of market fragmentation on real investment decisions. By lowering trading costs, market fragmentation may encourage information acquisition and trade on such information, leading to more informative stock prices. More informative stock prices can affect real capital investment for at least two reasons. First, managers can learn about their own firms' investment opportunities with greater precision. Second, prices can reveal information to creditors that reduces uncertainty about the firm's investment opportunities or information asymmetry between creditors and borrowing firms. This reduction in uncertainty and/or information asymmetry can alleviate financing constraints that limit the pursuit of profitable investment opportunities. In either case, we predict that market fragmentation leads to an increase in the sensitivity of capital investment to investment opportunities.

To test our prediction, following prior literature, we estimate a regression of capital investment on Tobin's q, a standard measure of investment opportunities, which we modify to include an interaction of this variable with market fragmentation. Findings reveal that, consistent with our prediction, the association between capital investment and investment opportunities is increasing in market fragmentation. The sensitivity of investment to investment opportunities increases by 6.8% across the interquartile range of market fragmentation.

A recent theoretical study, Dávila and Parlatore (2021), shows that a decrease in trading costs can increase price informativeness if investors are heterogeneous in the precision of private information about the firm's fundamental value. Therefore, we estimate an augmented version of our capital investment equation that permits the interaction between investment opportunities and market fragmentation to vary with a proxy for investor heterogeneity, the concentration of investors across investor types. Findings reveal that the effect of market fragmentation on the sensitivity of investment to investment opportunities is greater when investors are more heterogeneous.

Next, we explore two mechanisms by which market fragmentation may influence real capital investment: (1) how it increases revelatory price efficiency and (2) who learns from more efficient prices, that is, managers or creditors. Regarding the first mechanism, because market fragmentation reflects increased competition among trading venues, investors may benefit through lower trading costs. If lower trading costs allow investors to realize greater benefits from informed trades, information acquisition and trading activity will increase. We investigate the extent to which market fragmentation spurs information acquisition by investors by regressing the number of downloads from the Electronic Data Gathering, Analysis, and Retrieval (EDGAR) system, our proxy for information acquisition, onto beginning-of-year market fragmentation and a set of control variables. We find that market fragmentation is associated with higher levels of EDGAR download activity, which is consistent with market fragmentation increasing incentives for information acquisition.

The increase in information acquisition associated with market fragmentation suggests that there should be a commensurate increase in informed trading. To test this prediction, we regress a measure of informed trading onto beginning-of-year market fragmentation and a set of control variables. We find that informed trading increases with market fragmentation. Based on our findings that information acquisition and informed trading increase with market fragmentation, we expect current stock prices will more fully reflect future earnings news. Consistent with this prediction, we find that firms with stocks that exhibit greater market fragmentation have higher future earnings response coefficients, that is, a higher association between current stock returns and future earnings.

Regarding the second mechanism, we next investigate the extent to which market fragmentation leads to prices that inform managers, that is, the managerial learning channel, or creditors, that is, the financing channel, about a firm's investment opportunities. We do this by comparing the effect of market fragmentation on the sensitivity of investment to investment opportunities for firms with differing levels of financing constraints. Because the association between market fragmentation and the sensitivity of investment to investment opportunities increases in financing constraints, the evidence suggests that the effects of market fragmentation work through the financing channel. Additional evidence indicates that the effect of market fragmentation on investment is weaker when firms finance investment primarily through private debt from creditors who likely have access to a richer set of information beyond stock prices. Notably, there is no effect of market fragmentation on investment for firms that rely primarily on private debt, suggesting the financing channel works through public debt. To provide additional evidence that the financing channel works through public debt, we analyze the relation between market fragmentation and the amount and yield of newly issued public debt. We find that fragmentation is associated with a higher sensitivity of new public debt to investment opportunities and with lower yields on new public debt issues, which is consistent with market fragmentation alleviating constraints for public debt financing.

We find evidence that the effects of market fragmentation also work through the managerial learning channel. In particular, we find that (a) the sensitivity of investment to investment opportunities increases with market fragmentation when financing constraints are low, and (b) the effect of market fragmentation on the sensitivity of investment to investment opportunities is lower for firms with executives who are more informed, that is, whose insider trades are more profitable. The latter finding suggests that market fragmentation facilitates less-informed managers in learning from prices.

One alternative explanation for a higher association between investment and investment opportunities when market fragmentation is high is that market fragmentation reduces noise in the measure of investment opportunities. That is, there is no change in firms' investment, but the empirical association between investment and investment opportunities is higher because there is less measurement error in investment opportunities. If this were the sole explanation for the higher association, we would not expect to observe real effects of market fragmentation. To examine this possibility, we test whether market fragmentation is associated with higher levels of future profitability, controlling for other factors. We find that market fragmentation is positively associated with future profitability (i.e., return on assets). We also find that the association between current capital expenditures and future profitability is higher when market fragmentation is higher, which is consistent with market fragmentation resulting in more profitable capital investment decisions.

We use two empirical approaches to address the possibility that factors causing firms to have more fragmented trading also are associated with the responsiveness of investment to investment opportunities. The first approach is a difference-in-differences analysis around the introduction of Regulation National Market System (Reg NMS) in 2007. Reg NMS requires that exchanges share data and that trades are executed at the best-quoted price across exchanges, and its introduction led to an increase in market fragmentation by removing some of the advantages held by larger, more established exchanges. The second approach uses the number of U.S. trading venues as an instrument for market fragmentation. Inferences based on these tests are consistent with those of our main analyses and generally consistent with those of our additional analyses. In particular, we find that the sensitivity of investment to investment opportunities, information gathering, and the incorporation of future earnings into stock price all increased significantly following Reg NMS, but there was no significant change in informed trading. Although these analyses help to address the possibility that confounding factors may explain our results, we acknowledge that limitations inherent to these tests prevent us from conclusively drawing causal inferences.

Taken together, our study's findings provide the first empirical evidence of real effects of market fragmentation on corporate decision making. Whereas prior research on market fragmentation documents short-term effects on trading costs, including bid-ask spreads and the speed with which prices reflect market information, we show that market fragmentation is also associated with more information gathering by investors, more informed trade, and greater incorporation of future earnings into prices, revelation of information to managers and to creditors, and, ultimately, more efficient capital investment decisions. Our results suggest that market fragmentation improves revelatory price efficiency, which informs managers and lenders about a firm's investment opportunities. Our research complements recent work on factors associated with revelatory price efficiency (Chen et al. 2007, Jayaraman and Wu 2019) by documenting a new factor that affects revelatory price efficiency, that is, equity market fragmentation; by providing empirical evidence of a mechanism through which revelatory price efficiency increases, that is, increased information acquisition; and by extending the notion of revelatory price efficiency from managers learning from prices to include public creditors as well.

2. Background and Predictions

2.1. Market Fragmentation

2.1.1. Markets and Regulation. Although most U.S. firms initially list their stocks on either the New York Stock Exchange (NYSE) or the Nasdaq Stock Market, their stocks may also be traded on a number of other exchanges and trading venues.[2] The number of U.S. stock exchanges has increased over the last two decades, from 8 in 1995 to 13 by the end of 2017. Each exchange has different features and listing requirements. For example, whereas the NYSE has a trading floor, other exchanges are entirely electronic. Some exchanges offer colocation services that allow investors, typically institutional investors, faster trade execution. Other exchanges deliberately introduce "speed bumps" that allow market makers to reprice bids and offers before an informed trade is executed. Exchanges compete for trading volume based on liquidity, cost, speed, and these other features.

Investors can choose which exchange to trade in. However, investors usually use "smart order routing" software that analyzes prices, liquidity, and features of various trading venues (e.g., speed, fees, etc.) to detect and execute the best available opportunity based on the conditions selected by the investor. Frequently, large orders are split into suborders that are executed across various trading venues.

As a result of the increase in the number of stock exchanges and technology to route orders efficiently to the various exchanges, the U.S. equity market increasingly exhibits a large degree of fragmentation, which is the extent to which equity trades are dispersed across multiple exchanges rather than concentrated in one or a small number of exchanges. Whereas into the 1990s most equity trading took place on the floor of the NYSE, today no single exchange holds even a 20% market share (see Appendix B for the distribution of trading volume across U.S. exchanges). Even further fragmentation in exchange volume is expected in the future (Redfearn 2019).

The proliferation of exchanges has been supported by the Securities and Exchange Commission (SEC), which issued two major rulings to increase competition among exchanges and promote the dissemination and sharing of information across exchanges. The first ruling, the Securities Acts Amendments of 1975 (U.S. Securities and Exchange Commission 1975), set up a national market system and a system for nationwide clearing and settlement of securities transactions. As a result, investors were provided with information to compare price quotes across trading venues, leading to a convergence in prices across venues.

More recently, the SEC passed Reg NMS (U.S. Securities and Exchange Commission 2005), which was designed to increase competition in equity markets, thus promoting more efficient trading services and pricing of stocks (SEC Release No. 34-61358).[3] A key provision of the regulation, the access rule, reinforces the sharing of market data across exchanges that was originally prescribed in the Securities Acts Amendments of 1975. Another key provision, the order protection rule or trade-through rule, requires exchanges to guarantee that trades are executed at prices no worse than those available at other trading venues. Together, the sharing of data and guarantee of order protection increased competition across exchanges by removing some of the advantages held by larger, more established exchanges. For example, newer or smaller exchanges can observe the quotes of larger exchanges and attract order flow by beating those quotes.[4]

There are at least two benefits to investors from increased fragmentation. First, by increasing competition among trading venues, fragmentation could lower average trading costs as exchanges offer favorable terms and lower fees to investors and innovate in ways that reduce latency. Second, the proliferation of trading venues could allow venues to specialize, offering different services to investors (e.g., colocation or access to data). On the other hand, fragmentation may increase investors' trading costs. Notably, as trading volume is divided across multiple trading venues, execution risk may increase as investors find it more difficult to find counterparties, particularly when trading in stocks with lower volume.[5] Whether fragmentation leads to a net cost or benefit to investors is an empirical question. As we discuss in the next section, empirical research documents that market fragmentation reduces trading costs and improves market quality, especially for larger firms.

2.1.2. Related Research. Theoretical research highlights both disadvantages and advantages to market fragmentation. Mendelson (1987) shows that fragmentation may isolate individuals for whom there are mutually beneficial trades because they are trading on different venues. Chowdhry and Nanda (1991) posits that fragmentation can increase adverse selection costs. Furthermore, Pagano (1989) predicts that stocks trading in less liquid markets may be adversely affected by fragmentation. As the set of potential counterparties is spread across more trading venues, it becomes more difficult for individual traders to find a counterparty in any particular venue.

On the positive side, Stoll (2001) suggests that the ability to direct trades to specific trading venues can foster price efficiency, as different exchanges offer unique features to satisfy traders' needs (Harris and Raviv 1993). Malamud and Rostek (2017) shows that when traders' risk preferences are sufficiently heterogeneous, fragmented markets produce outcomes that are welfare superior to a centralized market. Chen and Duffie (2021) shows that fragmentation leads to more aggressive order submission (across all exchanges) and better rebalancing of unwanted positions across traders, which leads to prices that better reflect traders' private information. This occurs because traders can split orders across exchanges to limit their price impact on a given exchange.

Empirical studies examine the effects of market fragmentation on various dimensions of market quality, including bid-ask spreads, trading costs, and the speed with which prices reflect market information. O'Hara and Ye (2011) finds that U.S. stocks with more fragmented trading have lower bid-ask spreads, faster execution of trades, and price movements that are closer to a random walk. The benefits of market fragmentation have also been documented for European equity markets (Foucault and Menkveld 2008, Degryse et al. 2015, Félez-Vinas 2017, Gresse 2017), Australian equity markets (Aitken et al. 2017), and U.S. option markets (De Fontnouvelle et al. 2003). An exception is Chung and Chuwonganant (2012), which finds that order execution speed is slower, order fill rate is lower, and order cancellation rate is higher for most trades after Reg NMS, which resulted in increased market fragmentation.

A few studies document that the effects of market fragmentation vary with the size of firms (O'Hara and Ye 2011, Degryse et al. 2015, Gresse 2017, Haslag and Ringgenberg 2023). For example, Haslag and Ringgenberg (2023) shows that the effects of fragmentation differ depending on market capitalization. Specifically, although the study finds an on-average negative association between fragmentation and bid-ask spreads (a one standard deviation increase in fragmentation is associated with a 2.2% reduction in bid-ask spreads), the association is significantly more negative for large firms but positive for small firms. Similarly, the association between fragmentation and the speed of price adjustment to new information is positive for large firms but negative for small firms. Haslag and Ringgenberg (2023) concludes that the negative effects of Reg NMS documented by Chung and Chuwonganant (2012) are concentrated in small firms.

2.2. Hypothesis Development

Academic research documents that managers incorporate investors' private information obtained from stock prices when making corporate investment decisions (Hayek 1945, Gebhardt et al. 2005, Chen et al. 2007, Downing et al. 2009, Bakke and Whited 2010, Jayaraman and Wu 2019, Ye et al. 2019). As explained in Chen et al. (2007), if at a given point in time managers decide on the level of investment attempting to maximize the expected value of the firm, they will use all information available to them at that point. This includes both the information in the stock price and other information that managers have and that has not yet been incorporated into the stock price. Indeed, a recent survey of Chinese public firms by Goldstein et al. (2021) indicates that 90% of managers pay attention to the stock market to learn information to guide investment decisions. In this case, investment will be more sensitive to stock price when the price provides more information that is new to managers. Bond et al. (2012) characterizes the extent to which stock prices reveal information that can be useful in investment decisions as "revelatory price efficiency" (RPE). As RPE increases, the sensitivity of capital investment to Tobin's q increases.

Although prior research provides evidence that equity market fragmentation lowers trading costs and thus improves market quality, prior research has not addressed whether equity market fragmentation leads to greater RPE. Dávila and Parlatore (2021) develops a theoretical model showing that a decrease in trading costs can increase price informativeness if investors are heterogeneous in the precision of private information about the firm's fundamental value. By lowering trading costs, market fragmentation can lead to greater opportunities for traders to benefit from informed trades, and thus we expect more information gathering and more information-based trades.[6] To the extent the conditions laid out in Dávila and Parlatore (2021) are met, that is, heterogeneity in the precision of private information, this will result in greater price informativeness and hence greater RPE.

If market fragmentation leads to prices with greater RPE, that is, prices that reveal with greater precision information to managers about their firms' investment opportunities, then we expect a higher association between capital investment and Tobin's q when market fragmentation is higher. This condition is met when equity market participants have information, revealed through prices, that managers do not have about the firm's investment opportunities. Managers' learning about their own investment opportunities from external sources does not imply that they know less than outsiders, but only that outsiders have some information that managers do not possess (Ferracuti and Stubben 2019). We refer to the mechanism by which managers learn from prices about their firms' investment opportunities as the managerial learning channel.

Prior research shows that creditors can learn from equity prices (Gebhardt et al. 2005, Downing et al. 2009). If stock prices inform the firm's external capital providers about the firm's investment opportunities, then stock prices will reduce financing constraints and facilitate capital investment.[7] One way that informative stock prices can facilitate capital investment is by reducing uncertainty about the value of the firm's investment opportunities. Because financial constraints are likely to be greater when creditors face higher uncertainty regarding a firm's future prospects, we expect higher uncertainty to be negatively associated with investment (Bo et al. 2003).

Another way that informative stock prices can facilitate capital investment is by reducing information asymmetry between the firm and its creditors, if creditors learn more from stock prices than managers do. When there is information asymmetry between a firm and its capital providers, the firm cannot credibly convey the value of its investment opportunities to them. This leads to financing constraints that limit otherwise profitable investments (Myers and Majluf 1984, Fazzari et al. 1988). In this setting of financing constraints, capital investment exhibits a lower association with marginal Tobin's q because not all investment opportunities can receive the necessary financing.

Thus, financing constraints can be mitigated if there are sources of information from outside the firm that reduce either information uncertainty or information asymmetry between managers and creditors, in particular for investors in public debt who do not have access to the richer set of information that private lenders do. If market fragmentation leads to prices that reveal information to creditors that reduces information uncertainty or information asymmetry between the creditors and borrowing firms, alleviating financing constraints that limit capital investment, we expect a higher association between capital investment and Tobin's q when market fragmentation is higher (Bond et al. 2004). We refer to the mechanism by which creditors learn from prices about borrowers' investment opportunities as the financing channel.

Regardless of whether the effects of market fragmentation work through the managerial learning channel or the financing channel, we hypothesize that an increase in market fragmentation leads to an increase in the sensitivity of capital investment to investment opportunities.[8]

3. Sample and Data

To construct the variables we use in our analyses, we obtain exchange location and trade data from the Trade and Quote (TAQ) database, stock price and return data from the Center for Research in Security Prices (CRSP), accounting data from Compustat, institutional ownership and insider trading data from Thomson Reuters, analyst coverage data from I/B/E/S, loan data from DealScan, bond data from the Mergent Fixed Income Securities Database, and data on EDGAR downloads from the SEC. We include only ordinary common shares in U.S. firms (share codes 10 and 11 in CRSP). We exclude observations with share prices below $1.00 and observations with missing values for the variables required in the regression analyses. We winsorize each continuous variable at the 1st and 99th percentiles. Our primary sample comprises 109,020 firm-year observations, with smaller subsamples based on data availability for some of our additional analyses.

Table 1, Panel A, lists the 14 trading venues in existence at the end of our sample period, 2017. The 14 trading venues include 13 exchanges and an aggregation of trades from various nonexchange trading venues (e.g., dark pools), which we obtain from the Financial Industry Regulatory Authority (FINRA).[9] Panel B lists, by year, the number of trading venues and sample mean of market fragmentation across those venues. The panel reveals two trends. First, the number of trading venues increases from 9 in 1995 to 15 in 2010 before leveling out at 14 by the end of the sample period. Second, fragmentation in trading across venues is generally increases throughout the sample period, rising from 0.08 in 1995 to 0.72 by 2017. The largest single-year increase occurs in 2008, the year following the implementation of Reg NMS, which was intended to increase competition among trading venues.

Table 1. Statistics on Market Fragmentation

Panel A: Trading venues as of December 31, 2017

Code	Trading venue name	Code	Trading venue name
A	NYSE MKT Stock Exchange	N	New York Stock Exchange
B	NASDAQ OMX BX Stock Exchange	P	NYSE Arca SM
C	National Stock Exchange	Q	NASDAQ Stock Exchange
D	Off-exchange Trading (from FINRA)	V	The Investors' Exchange
J	Direct Edge A Stock Exchange	X	NASDAQ OMX PSX Stock Exchange
K	Direct Edge X Stock Exchange	Y	BATS Y-Exchange
M	Chicago Stock Exchange	Z	BATS Exchange

Panel B: Market fragmentation by year

Year	Number of trading venues	Mean of *FRAG*	Year	Number of trading venues	Mean of *FRAG*
1995	9	0.08	2007	11	0.37
1996	9	0.08	2008	11	0.53
1997	9	0.08	2009	11	0.60
1998	9	0.08	2010	15	0.63
1999	9	0.09	2011	14	0.66
2000	9	0.11	2012	14	0.70
2001	9	0.11	2013	14	0.70
2002	10	0.10	2014	14	0.69
2003	10	0.13	2015	13	0.70
2004	10	0.26	2016	14	0.71
2005	10	0.36	2017	14	0.72
2006	11	0.35			

Notes. Table 1 reports data on market fragmentation between 1995 to 2017. Panel A displays the trading venues contained in the TAQ database as of December 31, 2017. Panel B reports the number of trading venues and sample mean of market fragmentation across those venues by year. The 14 trading venues include 13 exchanges and an aggregation of trades from various nonexchange trading venues (e.g., dark pools), which we obtain from FINRA. Market fragmentation, *FRAG*, equals one minus the Herfindahl index of trade value for every stock executed in each trading venues and ranges from zero (no fragmentation) to one (high fragmentation).

Table 2, Panels A and B, presents sample summary statistics and correlations among variables. Panel A reveals that the average firm invests 5.69% of total assets per year, has a Tobin's q of 2.21, and cash flows equal to 3.33% of total assets. Panel B reveals that market fragmentation, *FRAG*, and the number of trading venues, *#VENUES*, are positively correlated (0.85), as expected. Furthermore, market fragmentation is associated with lower levels of capital investment and lower values of Tobin's q; however, pairwise correlations cannot speak to the association between market fragmentation and the sensitivity of investment to Tobin's q, which is the focus of our regression analyses that follow.

To gain insight into the firm characteristics associated with market fragmentation, we estimate a descriptive regression of market fragmentation onto a set of firm-level characteristics. In particular, we estimate the following regression:

$$\begin{aligned} FRAG_{it} &= b_0 + b_1 SIZE_{it} + b_2 MB_{it} + b_3 ROA_{it} + b_4 LEVERAGE_{it} \\ &\quad + b_5 IDIOVOL_{it} + b_6 TURNOVER_{it} + b_7 ANALYST_{it} \\ &\quad + b_8 INSTOWN_{it} + b_9 NYSE_{it} + b_{10} NASDAQ_{it} \\ &\quad + Industry\ Fixed\ Effects + Year\ Fixed\ Effects + e_{it}, \end{aligned} \tag{1}$$

where *FRAG* equals one minus the Herfindahl index of the concentration of trading across trading venues during the fiscal year. Specifically, for each stock and fiscal year, we use the TAQ database to calculate *FRAG* as $1 - \Sigma\, s_j^{\,2}$, where s_j is the proportion of trades that occur on trading venue j.

SIZE is the natural log of total assets; *MB* is the ratio of market value of equity to book value of equity; *ROA* is net income before extraordinary items divided by total assets; *LEVERAGE* is the book value of total debt divided by sum of the book value of total debt and market value of equity; *IDIOVOL* is the standard deviation of idiosyncratic daily stock returns over the fiscal year, calculated from the Carhart (1997) four-factor model; *TURNOVER* is the natural log of trading volume during the fiscal year scaled by number of shares outstanding; *ANALYST* is the natural log of one plus the number of analysts who issue an earnings forecast during the fiscal year; and *INSTOWN* is the percentage of common shares outstanding held by institutional investors. We include indicator variables for the primary listing exchange of the firm's stock: the NYSE (*NYSE*), Nasdaq (*NASDAQ*), or any other exchange (indictor omitted). Appendix A provides definitions for all variables used in this study.

Column (1) of Table 2, Panel C, presents the results from the regression of *FRAG* onto firm characteristics

Table 2. Descriptive Statistics

Panel A: Summary statistics

Variable	N	Mean	Standard deviation	25%	Median	75%
$FRAG_{i,t-1}$	109,020	0.35	0.29	0.03	0.30	0.64
$CAPX_{it}$	109,020	5.69	7.96	1.01	3.16	7.01
$Q_{i,t-1}$	109,020	2.21	2.33	1.10	1.49	2.36
CF_{it}	109,020	3.33	18.05	0.41	5.46	11.68
$RETURN3_{it}$	109,020	0.03	1.10	−0.60	−0.12	0.37
$INVASSETS_{i,t-1}$	109,020	15.03	36.02	0.50	2.42	11.51
$SIZE_{it}$	109,020	6.18	2.22	4.56	6.13	7.68
MB_{it}	109,020	0.03	0.04	0.01	0.02	0.03
ROA_{it}	109,020	−0.02	0.20	−0.02	0.03	0.07
$LEVERAGE_{it}$	109,020	0.24	0.24	0.02	0.17	0.39
$IDIOVOL_{it}$	109,020	0.03	0.02	0.02	0.03	0.04
$TURNOVER_{it}$	109,020	0.47	0.37	0.20	0.39	0.65
$ANALYSTS_{it}$	109,020	1.49	1.12	0.00	1.61	2.40
$INSTOWN_{it}$	109,020	0.46	0.34	0.15	0.44	0.74
$NYSE_{it}$	109,020	0.38	0.49	0.00	0.00	1.00
$NASDAQ_{it}$	109,020	0.55	0.50	0.00	1.00	1.00
$EDGAR\_10KQ_{it}$	56,785	2.18	1.46	1.29	1.84	2.76
$EDGAR\_ALL_{it}$	56,785	3.16	1.62	2.01	2.58	3.63
PIN_EKO_{it}	79,448	0.22	0.13	0.13	0.19	0.28
PIN_OWR_{it}	28,766	0.46	0.26	0.24	0.47	0.66
$RETURN_{it}$	109,020	0.16	0.69	−0.21	0.06	0.35
$\Delta EARN_{it}$	109,020	0.00	0.15	−0.01	0.00	0.01
$\#VENUES_{it}$	109,020	10.56	1.85	9.00	10.00	12.25

Panel B: Pearson correlations among select variables

	Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
1	$FRAG_{i,t-1}$	1.00													
2	$CAPX_{it}$	−0.13	1.00												
3	$Q_{i,t-1}$	−0.08	0.16	1.00											
4	CF_{it}	0.06	0.16	−0.17	1.00										
5	$RETURN3_{it}$	−0.02	−0.04	−0.06	0.11	1.00									
6	$SIZE_{it}$	0.44	−0.05	−0.18	0.29	0.03	1.00								
7	MB_{it}	−0.01	0.06	0.33	−0.06	−0.05	−0.07	1.00							
8	ROA_{it}	0.07	0.05	−0.19	0.75	0.08	0.33	−0.08	1.00						
9	$LEVERAGE_{it}$	0.05	−0.04	−0.32	0.02	0.05	0.32	−0.23	0.04	1.00					
10	$IDIOVOL_{it}$	−0.37	0.05	0.15	−0.32	−0.01	−0.59	0.02	−0.43	−0.02	1.00				
11	$TURNOVER_{it}$	0.24	0.10	0.24	−0.02	−0.06	0.19	0.12	−0.08	−0.11	0.11	1.00			
12	$ANALYSTS_{it}$	0.35	0.08	0.12	0.21	0.02	0.58	0.09	0.17	−0.08	−0.37	0.40	1.00		
13	$INSTOWN_{it}$	0.33	−0.01	0.01	0.23	0.04	0.45	0.04	0.21	−0.05	−0.40	0.32	0.55	1.00	
14	$\#VENUES_{it}$	0.85	−0.14	−0.08	0.01	−0.02	0.31	−0.01	0.03	0.02	−0.30	0.17	0.22	0.20	1.00

Panel C: Firm-level characteristics associated with market fragmentation

	Dependent variable = $FRAG_{it}$	
Variable	(1)	(2)
$SIZE_{it}$	0.007***	0.012***
	(12.44)	(9.50)
MB_{it}	0.000	−0.034***
	(0.02)	(−3.39)
ROA_{it}	−0.019***	0.005
	(−7.56)	(1.53)
$LEVERAGE_{it}$	0.004	0.030***
	(1.31)	(7.06)
$IDIOVOL_{it}$	−0.006***	−0.007***
	(−15.84)	(−20.49)
$TURNOVER_{it}$	0.039***	0.036***
	(21.99)	(17.03)
$ANALYSTS_{it}$	0.016***	0.008***
	(19.14)	(6.46)

Table 2. (Continued)

Panel C: Firm-level characteristics associated with market fragmentation

Variable	Dependent variable = $FRAG_{it}$ (1)	(2)
$INSTOWN_{it}$	0.026***	0.010***
	(9.80)	(3.24)
$NYSE_{it}$	−0.035***	−0.025***
	(−8.87)	(−4.12)
$NASDAQ_{it}$	−0.097***	−0.108***
	(−26.48)	(−18.98)
Fixed effects	Industry, year	Firm, year
No. of observations	108,977	108,977
Adjusted R^2	0.879	0.916

Notes. Table 2 presents descriptive statistics for the sample. Panel A presents summary statistics, Panel B presents Pearson correlations among variables, and Panel C reports pooled OLS regression results from the estimation of Equation (1)—a regression of market fragmentation onto various firm characteristics. t statistics based on standard errors clustered by firm are shown in parentheses. Variable definitions are provided in Appendix A.

***, **, and *Significance at the 0.01, 0.05, and 0.10 levels, respectively.

with industry and year fixed effects. Firms that exhibit a greater degree of market fragmentation tend to be larger (*SIZE* coefficient = 0.007, t statistic = 12.44), less profitable (*ROA* coefficient = −0.019, t statistic = −7.56), and less volatile (*IDIOVOL* coefficient = −0.006, t statistic = −15.84). Firms with greater market fragmentation are more actively traded (*TURNOVER* coefficient = 0.039, t statistic = 21.99), are more actively followed by analysts (*ANALYST* coefficient = 0.016, t statistic = 19.14), and have higher levels of institutional ownership (*INSTOWN* coefficient = 0.026, t statistic = 9.80). Finally, firms with a primary listing on the NYSE (*NYSE* coefficient = −0.035, t statistic = −8.87) or NASDAQ (*NASDAQ* coefficient = −0.097, t statistic = −26.48) exhibit less market fragmentation than other firms.

Column (2) presents the results from the regression of *FRAG* onto firm characteristics with firm rather than industry fixed effects. When focusing on within-firm rather than across-firm variation, the results yield similar inferences with the following exceptions. Market fragmentation is higher in years when firms have a lower book-to-market ratio (*MB* coefficient = −0.034, t statistic = −3.39) and are more highly levered (*LEVERAGE* coefficient = 0.030, t statistic = 7.06).

4. Market Fragmentation and the Sensitivity of Investment to Investment Opportunities

To investigate the effect of fragmentation on capital investment, we adapt the investment model in Chen et al. (2007), which allows the sensitivity of investment to investment opportunities to vary with the informativeness of stock price. In particular, we estimate the following regression model, which permits the sensitivity of investment to investment opportunities to vary with the degree of market fragmentation:

$$\begin{aligned} CAPX_{it} &= b_0 + b_1 Q_{i,t-1} + b_2 Q_{i,t-1} \times FRAG_{i,t-1} \\ &\quad + b_3 FRAG_{i,t-1} + b_4 CF_{it} + b_5 RETURN3_{it} \\ &\quad + b_6 INVASSETS_{i,t-1} + \mathit{Firm\ Fixed\ Effects} \\ &\quad + \mathit{Year\ Fixed\ Effects} + e_{it}, \end{aligned} \tag{2a}$$

where *CAPX* is annual capital expenditures scaled by beginning-of-year total assets;[10] Q is the sum of the market value of equity and book value of debt, divided by the beginning-of-year book value of total assets; and *CF* is net operating cash flow, scaled by beginning-of-year total assets.[11] Following Chen et al. (2007), we include a control for the value-weighted, market-adjusted stock return over the current and subsequent two years, *RETURN3*, because prior research suggests firms tend to invest when their stocks are overvalued (Loughran and Ritter 1995, Baker and Wurgler 2002). Also, we include the inverse of total assets at the beginning of the year, *INVASSETS*, as a control for the spurious correlation between capital expenditures and Tobin's q that may arise from scaling capital expenditures by total assets (Chen et al. 2007). We include firm and year fixed effects as controls for unobserved heterogeneity across firms and over time, and we cluster standard errors by firm.

We predict b_2 is positive if market fragmentation is associated with an increase in revelatory price efficiency that informs either managers or creditors about the firm's investment opportunities.

Table 3 presents findings from estimations of Equations (2a) and (2b) in columns (2) and (3), respectively. Column (1) presents findings from a base model that

Table 3. Fragmentation and the Sensitivity of Investment to Investment Opportunities

	Dependent variable = $CAPX_{it}$		
Variable	(1)	(2)	(3)
$Q_{i,t-1}$	0.588***	0.548***	0.480***
	(24.41)	(20.05)	(9.78)
$Q_{i,t-1} \times FRAG_{i,t-1}$		0.248***	0.177
		(3.45)	(1.64)
$Q_{i,t-1} \times FRAG_{i,t-1} \times RKINVESTHET_{i,t-1}$			0.434***
			(2.41)
$Q_{i,t-1} \times RKINVESTHET_{i,t-1}$			−0.170**
			(−2.05)
$FRAG_{i,t-1} \times RKINVESTHET_{i,t-1}$			0.520
			(1.01)
$FRAG_{i,t-1}$		−2.888***	−2.406***
		(−8.63)	(−5.94)
$RKINVESTHET_{i,t-1}$			−1.084***
			(−4.00)
CF_{it}	0.033***	0.032***	0.032***
	(10.12)	(9.93)	(9.60)
$RETURN3_{it}$	−0.294***	−0.301***	−0.288***
	(−10.69)	(−11.00)	(−9.70)
$INVASSETS_{i,t-1}$	0.021***	0.021***	0.039***
	(8.38)	(8.33)	(10.60)
Firm and year fixed effects	Yes	Yes	Yes
No. of observations	109,020	109,020	81,320
Adjusted R^2	0.591	0.592	0.596

Notes. Table 3 reports pooled OLS regression results from the estimation of Equations (2a) and (2b). *CAPX* is capital expenditures as a percentage of total assets. *Q* is Tobin's q. *CF* is cash flow from operations as a percentage of total assets. *FRAG* equals one minus the Herfindahl index of the concentration of trading across trading venues during the fiscal year. *RKINVESTHET* equals one minus the Herfindahl index of the proportion of holdings of various investor types defined in Bushee (1998): noninstitutional investors and transient, quasi-indexer, and dedicated institutional investors, ranked into deciles and scaled between zero and one. All other variables are defined in Appendix A. *t*-statistics based on standard errors clustered by firm are shown in parentheses.

***, **, and *Significance at the 0.01, 0.05, and 0.10 levels, respectively.

excludes market fragmentation. Consistent with prior research, the base model estimation indicates that capital investment is significantly positively associated with investment opportunities (*Q* coefficient = 0.588, *t* statistic = 24.41) and internal cash flows (*CF* coefficient = 0.033, *t* statistic = 10.12).[12]

Consistent with our prediction, the findings in column (2) reveal that the association between capital investment and investment opportunities is increasing in market fragmentation. In particular, the coefficient on the interaction between *Q* and *FRAG* is significantly positive (coefficient = 0.248, *t* statistic = 3.45).[13] In terms of economic significance, based on the average within-firm distribution of *FRAG* (deHaan 2021), the sensitivity of investment to investment opportunities increases from 0.600 (=0.548 + 0.209 × 0.248) at the first quartile of *FRAG* to 0.641 (=0.548 + 0.374 × 0.248) at the third quartile of *FRAG*, which is an increase of 6.8%.[14]

As noted previously, Dávila and Parlatore (2021) shows that a decrease in trading costs can increase price informativeness if investors are heterogeneous in the precision of private information about the firm's fundamental value. Therefore, we estimate a version of Equation (2a) that allows the effect of market fragmentation to vary with investor heterogeneity.

$$\begin{aligned} CAPX_{it} &= b_0 + b_1 Q_{i,t-1} + b_2 Q_{i,t-1} \times FRAG_{i,t-1} \\ &\quad + b_3 Q_{i,t-1} \times FRAG_{i,t-1} \times RKINVESTHET_{i,t-1} \\ &\quad + b_4 Q_{i,t-1} \times RKINVESTHET_{i,t-1} \\ &\quad + b_5 FRAG_{i,t-1} \times RKINVESTHET_{i,t-1} + b_6 FRAG_{i,t-1} \\ &\quad + b_7 RKINVESTHET_{i,t-1} + b_8 CF_{it} + b_9 RETURN3_{it} \\ &\quad + b_{10} INVASSETS_{i,t-1} + Firm\ Fixed\ Effects \\ &\quad + Year\ Fixed\ Effects + e_{it} \end{aligned} \tag{2b}$$

We measure investor heterogeneity, *RKINVESTHET*, as one minus a Herfindahl index of the proportion of holdings of various investor types defined in Bushee (1998): noninstitutional investors and transient, quasi-indexer, and dedicated institutional investors. *RKINVESTHET* is ranked into deciles and scaled between zero and one. We predict b_3 is positive if the effects of market fragmentation increase with investor heterogeneity.

The findings in Table 3, column (3), reveal that, as expected, the effects of market fragmentation on the sensitivity of investment to investment opportunities increase with investor heterogeneity. In particular, the coefficient on the three-way interaction among Q, *FRAG*, and *RKINVESTHET* is significantly positive (coefficient = 0.660, t statistic = 3.49).

5. How Market Fragmentation Improves Revelatory Price Efficiency

Next, we examine one potential mechanism by which market fragmentation leads to prices with greater revelatory efficiency. Prior research suggests that market fragmentation results in lower trading costs. The decrease in trading costs may result in an increase in information acquisition and thus more informed trading and informative prices.

To test whether market fragmentation results in an increase in information acquisition, we estimate the following regression model:

$$\begin{aligned} &EDGAR_{it}\\ &= b_0 + b_1 FRAG_{i,t-1} + b_2 SIZE_{it} + b_3 MB_{it} + b_4 ROA_{it}\\ &\quad + b_5 LEVERAGE_{it} + b_6 IDIOVOL_{it} + b_7 TURNOVER_{it}\\ &\quad + b_8 ANALYST_{it} + b_9 INSTOWN_{it} + Firm\ Fixed\ Effects\\ &\quad + Year\ Fixed\ Effects + e_{it}, \end{aligned} \tag{3}$$

where *EDGAR* is the natural log of the number of downloads during the year from the SEC's EDGAR database of either 10-K/10-Q filings (*EDGAR_10KQ*) or any filing (*EDGAR_ALL*). Because prior research shows that investors and other market participants access firms' regulatory filings to inform investment decisions (Drake et al. 2015, Lee et al. 2015, Loughran and McDonald 2017, Chen et al. 2020, Gibbons et al. 2020, Crane et al. 2023), we use EDGAR downloads as a proxy for information acquisition.[15] Table 4, Panel A, which presents the findings from the estimation of Equation (3), reveals that when

Table 4. How Market Fragmentation Improves Revelatory Price Efficiency

Panel A: EDGAR downloads		
Variable	(1) Dependent variable = #$EDGAR\_10KQ_t$	(2) Dependent variable = #$EDGAR\_ALL_t$
$FRAG_{i,t-1}$	0.114***	0.150***
	(2.69)	(5.93)
$SIZE_{it}$	0.116***	0.153***
	(8.97)	(22.75)
MB_{it}	0.129*	0.063
	(1.70)	(1.20)
ROA_{it}	−0.116***	−0.248***
	(−3.99)	(−14.18)
$LEVERAGE_{it}$	−0.098***	−0.076***
	(−2.83)	(−3.66)
$IDIOVOL_{it}$	0.204	1.291***
	(0.69)	(6.69)
$TURNOVER_{it}$	0.187***	0.271***
	(8.47)	(23.57)
$ANALYSTS_{it}$	0.021*	−0.015***
	(1.93)	(−2.61)
$INSTOWN_{it}$	0.088***	−0.023*
	(4.49)	(−1.86)
Firm and year fixed effects	Yes	Yes
No. of observations	56,785	56,785
Adjusted R^2	0.918	0.970
Panel B: Informed trading		
Variable	(1) Dependent variable = PIN_EKO_{it}	(2) Dependent variable = PIN_OWR_{it}
$FRAG_{i,t-1}$	0.014***	0.369***
	(3.66)	(11.73)
$SIZE_{it}$	−0.029***	−0.016***
	(−29.11)	(−3.75)
$INVPRICE_{it}$	0.108***	0.248***
	(28.76)	(9.77)
Firm and year fixed effects	Yes	Yes

Table 4. (Continued)

Panel B: Informed trading

Variable	(1) Dependent variable $= PIN\_EKO_{it}$	(2) Dependent variable $= PIN\_OWR_{it}$
No. of observations	79,448	28,766
Adjusted R^2	0.672	0.325

Panel C: Future earnings response coefficients

Variable	(1) Dependent variable $= RETURN_{it}$
$\Delta EARN_{it}$	0.383***
	(4.29)
$\Delta EARN_{i,t+1}$	−0.042
	(−1.38)
$\Delta EARN_{it} \times FRAG_{i,t-1}$	0.035
	(0.25)
$\Delta EARN_{i,t+1} \times FRAG_{i,t-1}$	0.228***
	(3.05)
$FRAG_{i,t-1}$	−0.202
	(−1.55)
$RETURN_{i,t+1}$	−0.092**
	(−2.45)
$RETURN_{i,t-1}$	−0.159***
	(−4.47)
$SIZE_{i,t-1}$	−0.279***
	(−7.18)
$MB_{i,t-1}$	−1.303***
	(−5.87)
Firm and year fixed effects	Yes
No. of observations	109,020
Adjusted R^2	0.196

Notes. Table 4 reports pooled OLS regression results from the estimation of Equations (3)–(5). *FRAG* equals one minus the Herfindahl index of the concentration of trading across trading venues during the fiscal year. In Panel A, *EDGAR* is the natural log of the number of downloads during the year from the SEC's EDGAR database of either 10-K/10-Q filings (*EDGAR_10KQ*) or any filing (*EDGAR_ALL*). In Panel B, *PIN* is the probability of informed trade, measured following Easley et al. (1996) (*PIN_EKO*) or following Odders-White and Ready (2008) (*PIN_OWR*). In Panel C, *RETURN* is the firm's buy-and-hold stock return over the fiscal year. $\Delta EARN$ is the change in net income before extraordinary items scaled by equity market value. All other variables are defined in Appendix A. t statistics based on standard errors clustered by firm are shown in parentheses.

***, **, and *Significance at the 0.01, 0.05, and 0.10 levels, respectively.

market fragmentation is higher, the firm's SEC filings are more frequently downloaded. This is true for 10-K and 10-Q filings ($FRAG_{it}$ coefficient $= 0.114$, t statistic $= 2.69$) and for all SEC filings ($FRAG_{it}$ coefficient $= 0.150$, t statistic $= 5.93$). As *FRAG* increases from the first quartile to the third quartile of its within-firm distribution, 10-K/10-Q downloads (all downloads) increase by 0.94% (0.89%) of the sample median value.

The increase in information acquisition associated with market fragmentation suggests that there should be a commensurate increase in informed trading. To test this prediction, we estimate the following regression model, which is adapted from Goldstein et al. (2020):

$$PIN_{it} = b_0 + b_1 FRAG_{i,t-1} + b_2 SIZE_{it} + b_3 INVPRICE_{it} + Firm\ Fixed\ Effects + Year\ Fixed\ Effects + e_{it}, \tag{4}$$

where *PIN* is the probability of informed trade, measured either following Easley et al. (1996) (*PIN_EKO*) or following Odders-White and Ready (2008) (*PIN_OWR*).[16] Following Goldstein et al. (2020), we include controls for firm size (*SIZE*) and the inverse of stock price (*INVPRICE*). Table 4, Panel B, which presents the findings from the estimation of Equation (4), reveals that the probability of informed trading increases with market fragmentation (*FRAG* coefficient $= 0.014$, t statistic $= 3.66$ in column (1); *FRAG* coefficient $= 0.369$, t statistic $= 11.73$ in column (2)). As *FRAG* increases from the first quartile to the third quartile of its within-firm distribution, *PIN_EKO* (*PIN_OWR*) increases by 1.02% (10.58%) of the sample median value.

Based on our findings that information acquisition and informed trading increase with market fragmentation, we expect that stock prices will more fully reflect information in future earnings. To test this conjecture, we augment a future-earnings-response-coefficient model (Lundholm and Myers 2002, Ettredge et al. 2005) with market fragmentation and its interactions with current and future earnings:

$$\begin{aligned} &RETURN_{it} \\ &= b_0 + b_1\Delta EARN_{it} + b_2\Delta EARN_{i,t+1} \\ &\quad + b_3\Delta EARN_{it} \times FRAG_{i,t-1} + b_4\Delta EARN_{i,t+1} \times FRAG_{i,t-1} \\ &\quad + b_5 FRAG_{i,t-1} + b_6 RETURN_{i,t+1} + b_7 RETURN_{i,t-1} \\ &\quad + b_8 SIZE_{i,t-1} + b_9 MB_{i,t-1} + \textit{Firm Fixed Effects} \\ &\quad + \textit{Year Fixed Effects} + e_{it}, \end{aligned} \tag{5}$$

where *RETURN* is the firm's annual buy-and-hold stock return over the fiscal year and $\Delta EARN$ is the change in income before extraordinary items, scaled by market value of equity at the beginning of the year. If market fragmentation leads to stock prices that more quickly incorporate future earnings information, we predict b_4 is positive. We include as controls proxies for risk factors identified in prior research. These include the prior-year stock return, firm size (*SIZE*), and the ratio of market value of equity to book value of equity (*MB*). We estimate Equation (4) with firm and year fixed effects.

Table 4, Panel C, presents findings from the estimation of Equation (5). When fragmentation equals zero, the contemporaneous earnings response coefficient is positive, which is consistent with prior research ($\Delta EARN_{it}$ coefficient = 0.383, t statistic = 4.29). This coefficient does not vary significantly with the degree of market fragmentation, that is, the $\Delta EARN_{it} \times FRAG$ coefficient, 0.035, is insignificant (t statistic = 0.25). However, the future earnings response coefficient increases as market fragmentation increases, that is, the $\Delta EARN_{i,t+1} \times FRAG$ coefficient, 0.228, is significantly positive (t statistic = 3.05). As *FRAG* increases from the first quartile to the third quartile of its within-firm distribution, the future earnings response coefficient increases from 0.006 to 0.043, which is consistent with the notion that market fragmentation leads to stock prices that more quickly incorporate future earnings information.

6. Does Fragmentation Work Through the Managerial Learning Channel, Financing Channel, or Both?

As described in Section 2.2, there are two channels through which market fragmentation can increase the sensitivity of capital investment to Tobin's q. Market fragmentation can lead to prices that reveal with greater precision information either to managers (the managerial learning channel) or to creditors (the financing channel) about the firm's investment opportunities. To determine the extent to which one or both the channels contributes to the increase in sensitivity of capital investment to investment opportunities, we estimate the following regression model, which augments Equation (2a) with a proxy for financing constraints:

$$\begin{aligned} &CAPX_{it} \\ &= b_0 + b_1 Q_{i,t-1} + b_2 Q_{i,t-1} \times FRAG_{i,t-1} \\ &\quad + b_3 Q_{i,t-1} \times FRAG_{i,t-1} \times RKCONSTR_{i,t-1} \\ &\quad + b_4 Q_{i,t-1} \times RKCONSTR_{i,t-1} \\ &\quad + b_5 FRAG_{i,t-1} \times RKCONSTR_{i,t-1} + b_6 FRAG_{i,t-1} \\ &\quad + b_7 RKCONSTR_{i,t-1} + b_8 RETURN3_{it} \\ &\quad + b_9 INVASSETS_{i,t-1} + \textit{Firm Fixed Effects} \\ &\quad + \textit{Year Fixed Effects} + e_{it}, \end{aligned} \tag{6}$$

where *RKCONSTR* is the decile rank of the delay investment score as measured by Hoberg and Maksimovic (2015), scaled between zero and one.[17] If the effects of market fragmentation work only through the managerial learning channel (financing channel), then we predict b_2 (b_3) is positive and b_3 (b_2) is zero. If the effects of market fragmentation work at least partially through both channels, we predict both b_2 and b_3 are positive.

The findings presented in Table 5, Panel A, reveal that the effects of market fragmentation on the sensitivity of investment to investment opportunities are larger for firms that are more financially constrained. In particular, the coefficient on the three-way interaction among Q, *FRAG*, and *RKCONSTR* is significantly positive (coefficient = 0.332, t statistic = 1.92), which indicates that the interaction between Q and *FRAG* increases from 0.202 to 0.534 across the deciles of *RKCONSTR*. This finding suggests that the effects of market fragmentation work through the financing channel, where market fragmentation benefits financially constrained firms that are more likely to rely on external financing. The positive coefficient on the interaction between Q and *FRAG* (i.e., when financing constraints are in the lowest decile) provides evidence that the effects of market fragmentation also work through the managerial learning channel (coefficient = 0.202, t statistic = 1.84).[18]

Another approach to provide evidence on the financing channel uses a firm's accounting quality. Firms can mitigate financing constraints by providing credible information to creditors. For example, Biddle and Hilary (2006) finds that better accounting information reduces information asymmetry between managers and outside suppliers of capital. Similarly, Biddle et al. (2009) finds that firms with better accounting quality are less likely to exhibit under-investment arising from financing

Table 5. Evidence for the Financing Channel

Panel A: Cross-sectional analysis based on the level of financing constraints

Variable	(1) Dependent variable = $CAPX_{it}$
$Q_{i,t-1}$	0.504***
	(9.71)
$Q_{i,t-1} \times FRAG_{i,t-1}$	0.202*
	(1.84)
$Q_{i,t-1} \times FRAG_{i,t-1} \times RKCONSTR_{i,t-1}$	0.332*
	(1.92)
$Q_{i,t-1} \times RKCONSTR_{i,t-1}$	−0.149**
	(−2.26)
$FRAG_{i,t-1} \times RKCONSTR_{i,t-1}$	−2.009***
	(−4.06)
$FRAG_{i,t-1}$	−1.688***
	(−4.53)
$RKCONSTR_{i,t-1}$	0.934***
	(4.07)
CF_{it}	0.034***
	(11.37)
$RETURN3_{it}$	−0.387***
	(−13.54)
$INVASSETS_{i,t-1}$	0.033***
	(10.34)
Firm and year fixed effects	Yes
No. of observations	51,383
Adjusted R^2	0.610

Panel B: Cross-sectional analysis based on accounting quality

Variable	(1) Dependent variable = $CAPX_{it}$
$Q_{i,t-1}$	0.600***
	(7.30)
$Q_{i,t-1} \times FRAG_{i,t-1}$	1.086***
	(5.23)
$Q_{i,t-1} \times FRAG_{i,t-1} \times RKAQ_{i,t-1}$	−1.280***
	(−5.12)
$Q_{i,t-1} \times RKAQ_{i,t-1}$	0.019
	(0.16)
$FRAG_{i,t-1} \times RKAQ_{i,t-1}$	1.421**
	(2.34)
$FRAG_{i,t-1}$	−3.523***
	(−6.95)
$RKAQ_{i,t-1}$	0.677**
	(2.20)
CF_{it}	0.038***
	(10.02)
$RETURN3_{it}$	−0.248***
	(−8.37)
$INVASSETS_{i,t-1}$	0.019***
	(5.85)
Firm and year fixed effects	Yes
No. of observations	72,808
Adjusted R^2	0.584

Panel C: Cross-sectional analysis based on the amount of private debt

Variable	(1) Dependent variable = $CAPX_{it}$
$Q_{i,t-1}$	0.853***
	(6.41)
$Q_{i,t-1} \times FRAG_{i,t-1}$	0.509*
	(1.94)

Table 5. (Continued)

Panel C: Cross-sectional analysis based on the amount of private debt

Variable	(1) Dependent variable = $CAPX_{it}$
$Q_{i,t-1} \times FRAG_{i,t-1} \times RKPRIVATE_{i,t-1}$	−0.582**
	(−2.04)
$Q_{i,t-1} \times RKPRIVATE_{i,t-1}$	0.054
	(0.33)
$FRAG_{i,t-1} \times RKPRIVATE_{i,t-1}$	1.813**
	(2.54)
$FRAG_{i,t-1}$	−1.543**
	(−2.32)
$RKPRIVATE_{i,t-1}$	−1.231***
	(−3.19)
CF_{it}	0.052***
	(11.09)
$RETURN3_{it}$	−0.277***
	(−6.91)
$INVASSETS_{i,t-1}$	0.042***
	(5.76)
Firm and year fixed effects	Yes
No. of observations	51,894
Adjusted R^2	0.681

Panel D: Debt amounts and debt yields

Variable	(1) Dependent variable = $DEBTAMT_t$	(2) Dependent variable = $DEBTYIELD_t$
Q_{t-1}	0.012	
	(0.97)	
$Q_{t-1} \times FRAG_{t-1}$	0.109***	
	(3.70)	
$FRAG_{t-1}$	−0.664***	−0.156**
	(−2.98)	(−2.14)
$SIZE_{t-1}$	0.346***	0.006
	(12.20)	(0.48)
ROA_{t-1}	0.001	−0.059
	(0.01)	(−0.75)
$LEVERAGE_{t-1}$	−0.007***	0.003***
	(−5.24)	(6.93)
$IDIOVOL_{t-1}$	2.975**	0.577
	(2.02)	(0.89)
$ANALYST_{t-1}$	−0.052*	−0.019*
	(−1.88)	(−1.65)
$INSTOWN_{t-1}$	−0.032	−0.055*
	(−0.44)	(−1.75)
Firm and year fixed effects	Yes	Yes
No. of observations	9,176	9,176
Adjusted R^2	0.632	0.622

Notes. Table 5 reports pooled OLS regression results from the estimation of Equations (6) through (9b). In Panels A and B, *CAPX* is capital expenditures as a percentage of total assets. *Q* is Tobin's *q*. *CF* is cash flow from operations as a percentage of total assets. *FRAG* equals one minus the Herfindahl index of the concentration of trading across trading venues during the fiscal year. *RKCONSTR* is the decile rank of the delay investment score (Hoberg and Maksimovic 2015), and *RKAQ* is the decile rank of the Dechow and Dichev (2002) earnings quality measure, as calculated by Francis et al. (2005). *RKCONSTR* and *RKAQ* are scaled between zero and one. In Panel C, *RKPRIVATE* is the decile rank of the ratio of private debt to total debt, scaled between zero and one. In Panel D, *DEBTAMT* is the natural log of the total dollar amount of bonds issued during the fiscal year, and *DEBTYIELD* is the weighted average yield on bonds issued during the year. All other variables are defined in Appendix A. *t* statistics based on standard errors clustered by firm are shown in parentheses.

***, **, and *Significance at the 0.01, 0.05, and 0.10 levels, respectively.

constraints. To the extent that accounting quality and market fragmentation are substitutes in reducing information asymmetry between firms and creditors, that is, the financing channel, the positive effect of market fragmentation on the sensitivity of capital investment to investment opportunities will be lower when accounting quality is higher.

To test our predictions, we estimate the following regression model:

$$\begin{aligned} CAPX_{it} &= b_0 + b_1 Q_{i,t-1} + b_2 Q_{i,t-1} \times FRAG_{i,t-1} \\ &\quad + b_3 Q_{i,t-1} \times FRAG_{i,t-1} \times RKAQ_{i,t-1} \\ &\quad + b_4 Q_{i,t-1} \times RKAQ_{i,t-1} + b_5 FRAG_{i,t-1} \times RKAQ_{i,t-1} \\ &\quad + b_6 FRAG_{t-1} + b_7 RKAQ_{i,t-1} + b_8 CF_{it} \\ &\quad + b_9 RETURN3_{it} + b_{10} INVASSETS_{it} \\ &\quad + Firm\ Fixed\ Effects + Year\ Fixed\ Effects + e_{it}, \end{aligned} \tag{7}$$

where *RKAQ* is the decile rank of the Dechow and Dichev (2002) earnings quality measure, calculated following Francis et al. (2005) and scaled between zero and one. If the effects of market fragmentation work through the financing channel, we predict b_3 is negative. If the effects of market fragmentation work only through the managerial learning channel, we predict b_3 equals zero.

The findings in Table 5, Panel B, reveal that the effects of market fragmentation on the sensitivity of investment to investment opportunities decreases with accounting quality. In particular, the coefficient on the three-way interaction among *Q*, *FRAG*, and *RKAQ* is significantly negative (coefficient = −1.280, *t* statistic = −5.12), which indicates that the interaction between *Q* and *FRAG* decreases from 1.086 to −0.194 across the deciles of *RKAQ*. As with the findings in Panel A, the findings in Panel B are consistent with market fragmentation working through the financing channel, where improved revelatory price efficiency resulting from market fragmentation and the quality of accounting information act as substitute sources of information to creditors.[19]

The findings in Panels A and B are consistent with market fragmentation providing information to creditors that alleviates financing constraints. However, this effect is likely to be smaller if the creditor has access to a richer set of information about the borrower, as is in the case of a private debt contract. Therefore, we predict the effect of market fragmentation on the sensitivity of investment to investment opportunities is decreasing in the extent of private debt as a fraction of total debt. To test this prediction, we estimate Equation (8):

$$\begin{aligned} CAPX_{it} &= b_0 + b_1 Q_{i,t-1} + b_2 Q_{i,t-1} \times FRAG_{i,t-1} \\ &\quad + b_3 Q_{i,t-1} \times FRAG_{i,t-1} \times RKPRIVATE_{i,t-1} \\ &\quad + b_4 Q_{i,t-1} \times RKPRIVATE_{i,t-1} \\ &\quad + b_5 FRAG_{i,t-1} \times RKPRIVATE_{i,t-1} + b_6 FRAG_{i,t-1} \\ &\quad + b_7 RKPRIVATE_{i,t-1} + b_8 CF_{it} + b_9 RETURN3_{it} \\ &\quad + b_{10} INVASSETS_{it} \\ &\quad + Firm\ Fixed\ Effects + Year\ Fixed\ Effects + e_{it}, \end{aligned} \tag{8}$$

where *RKPRIVATE* is the decile rank of the ratio of private debt to the sum of private debt and corporate bonds, using data from Capital IQ and scaled between zero and one. If the effects of market fragmentation are decreasing in the extent of private debt, then we predict b_3 is negative. The findings in Table 5, Panel C, are consistent with our prediction. In particular, the coefficient on the three-way interaction among *Q*, *FRAG*, and *RKPRIVATE* is significantly negative (coefficient = −0.582, *t* statistic = −2.04), which indicates that the interaction between *Q* and *FRAG* decreases from 0.509 to −0.073 across the deciles of *RKPRIVATE*. The sum of the coefficients on (a) the two-way interaction between *Q* and *FRAG* and (b) the three-way interaction among *Q*, *FRAG*, and *RKPRIVATE*, −0.073, indicates that the effect of market fragmentation on investment is not significantly different from zero (*F* statistic = 0.22, $p = 0.64$) for firms that rely primarily on private debt, suggesting the financing channel works through public debt.

To provide additional evidence that the financing channel works through public debt, we next analyze the relation between market fragmentation and the amount and yield of newly issued public debt. The demand for borrowing increases with the value of investment opportunities a firm faces. If market fragmentation eases the constraints associated with public debt financing, we expect that the association between new borrowings and investment opportunities increases in the degree of market fragmentation. When financing constraints are lower, firms are more freely able to borrow as needed to fund new investments. Furthermore, if market fragmentation reduces information asymmetry between borrowers and lenders, we expect that the association between yields on new public debt borrowings and the degree of market fragmentation is negative.

To test these predictions, we estimate the following regression models:

$$
\begin{aligned}
&DEBTAMT_t \\
&= b_0 + b_1Q_{t-1} + b_2Q_{t-1} \times FRAG_{t-1} + b_3FRAG_{t-1} \\
&\quad + b_4LEVERAGE_{t-1} + b_5SIZE_{t-1} + b_6ROA_{t-1} \\
&\quad + b_7IDIOVOL_{t-1} + b_8INSTOWN_{t-1} \\
&\quad + b_9ANALYST_{t-1} + Firm\ Fixed\ Effects \\
&\quad + Year\ Fixed\ Effects + e_t, \qquad (9a)
\end{aligned}
$$

$$
\begin{aligned}
&DEBTYIELD_t \\
&= b_0 + b_1FRAG_{t-1} + b_2LEVERAGE_{t-1} + b_3SIZE_{t-1} \\
&\quad + b_4ROA_{t-1} + b_5IDIOVOL_{t-1} + b_6INSTOWN_{t-1} \\
&\quad + b_7ANALYST_{t-1} + Firm\ Fixed\ Effects \\
&\quad + Year\ Fixed\ Effects + e_t, \qquad (9b)
\end{aligned}
$$

where *DEBTAMT* is the natural log of the total dollar amount of bonds issued during the fiscal year, and *DEBTYIELD* is the weighted average yield on bonds issued during the year. Each is calculated using data from the Mergent Fixed Income Securities Database. As defined previously, *Q* represents investment opportunities at the beginning of the fiscal year, and *FRAG* is the degree of market fragmentation at the beginning of the fiscal year. If market fragmentation reduces the constraints associated with public debt financing, then we predict b_2 is positive in Equation (9a) and b_1 is negative in Equation (9b).

Following prior studies (Blankespoor et al. 2013), we include controls for leverage, *LEVERAGE*, because firms with greater amounts of debt outstanding are not as freely able to borrow. All else equal, we expect that information asymmetry is lower between creditors and large, profitable borrowers with lower levels of equity volatility and higher levels of institutional ownership and analyst following, which results in greater access to credit for such borrowers. We measure firm size as the natural log of total assets, *(SIZE)*, and profitability as the return on assets, *ROA. IDIOVOL* is idiosyncratic volatility, *INSTOWN* is the percentage of the firm's shares that are held by institutions, and *ANALYST* is the natural log of one plus the number of analysts following the firm.

Table 5, Panel D, presents the results from the estimation of Equations (9a) and (9b). The significantly positive coefficient on the interaction between *Q* and *FRAG* in column (1) reveals that market fragmentation is associated with a higher sensitivity of new public debt to investment opportunities on average (coefficient = 0.109, *t* statistic = 3.70). In addition, the significantly negative coefficient on *FRAG* in column (2) reveals that market fragmentation is associated with lower average yields on new public debt issues (coefficient = −0.156, *t* statistic = −2.14).[20] As *FRAG* increases from the first quartile to the third quartile of its within-firm distribution, the sensitivity of *DEBTAMT* to investment opportunities increases by 34% and *DEBTYIELD* decreases by 1.22% of the sample median value. These findings are consistent with market fragmentation alleviating constraints for public debt financing.[21]

Taken together, the results presented in Table 5 suggest that the effects of market fragmentation work through the public-debt financing channel, where a firm's stock price decreases information asymmetry between the firm and public creditors by revealing information to creditors, which alleviates financing constraints.

The results presented in Table 5, Panel A, suggest that the effects of market fragmentation also work through the managerial learning channel. We explore this possibility further by examining the profitability of insider trades made by the firm's executives. We expect that managers are less likely to learn from stock prices when they already possess more private information about their firms. We measure managers' private information using the profitability of their insider trading, because insider trades reveal private, firm-specific information not impounded in price (Meulbroek 1992; Seyhun 1992, 1998; Damodoran and Liu 1993; Ke et al. 2002; Piotroski and Roulstone 2005; Chen et al. 2007). We estimate the following regression model:

$$
\begin{aligned}
&CAPX_{it} \\
&= b_0 + b_1Q_{i,t-1} + b_2Q_{i,t-1} \times FRAG_{i,t-1} \\
&\quad + b_3Q_{i,t-1} \times FRAG_{i,t-1} \times RKINSTRADE_{i,t-1} \\
&\quad + b_4Q_{i,t-1} \times RKINSTRADE_{i,t-1} \\
&\quad + b_5FRAG_{i,t-1} \times RKINSTRADE_{i,t-1} + b_6FRAG_{i,t-1} \\
&\quad + b_7RKINSTRADE_{i,t-1} + b_8CF_{it} + b_9RETURN3_{it} \\
&\quad + b_{10}INVASSETS_{i,t-1} + Firm\ Fixed\ Effects \\
&\quad + Year\ Fixed\ Effects + e_{it}, \qquad (10)
\end{aligned}
$$

where *RKINSTRADE* is the decile rank of insider trading profitability and scaled between zero and one. Following Jagolinzer et al. (2011) and Foucault and Fresard (2014), we measure the profitability of insiders' trades each day as the intercept (or alpha) from the four-factor Carhart (1997) model estimated over the 30 days following each transaction. We multiply the alpha by −1 for sales and calculate the average trading profits for each firm each year. If the effects of market fragmentation work through the managerial learning channel, then we predict b_3 is negative.

Table 6. Evidence for the Managerial Learning Channel

Variable	(1) Dependent variable = $CAPX_{it}$
$Q_{i,t-1}$	0.470***
	(15.44)
$Q_{i,t-1} \times FRAG_{i,t-1}$	0.391***
	(4.39)
$Q_{i,t-1} \times FRAG_{i,t-1} \times RKINSTRADE_{i,t-1}$	−0.363***
	(−2.59)
$Q_{i,t-1} \times RKINSTRADE_{i,t-1}$	0.261***
	(3.76)
$FRAG_{i,t-1} \times RKINSTRADE_{i,t-1}$	1.393***
	(4.10)
$FRAG_{i,t-1}$	−3.521***
	(−9.55)
$RKINSTRADE_{i,t-1}$	−0.873***
	(−5.35)
CF_{it}	0.033***
	(10.10)
$RETURN3_{it}$	−0.300***
	(−10.94)
$INVASSETS_{i,t-1}$	0.020***
	(8.01)
Firm and year fixed effects	Yes
No. of observations	109,020
Adjusted R^2	0.592

Notes. Table 6 reports pooled OLS regression results from the estimation of Equation (9). *CAPX* is capital expenditures as a percentage of total assets. *Q* is Tobin's *q*. *CF* is cash flow from operations as a percentage of total assets. *FRAG* equals one minus the Herfindahl index of the concentration of trading across trading venues during the fiscal year. *RKINSTRADE* is the decile rank of the profitability of insider trades made by the firm's executives during the year, scaled between zero and one. All other variables are defined in Appendix A. *t* statistics based on standard errors clustered by firm are shown in parentheses.

***, **, and *Significance at the 0.01, 0.05, and 0.10 levels, respectively.

The findings in Table 6 reveal that the effects of market fragmentation on the sensitivity of investment to investment opportunities decreases with the profitability of trading by the firm's insiders, which suggests the effects of market fragmentation may also work through the managerial learning channel. The coefficient on the three-way interaction among *Q*, *FRAG*, and *RKINSTRADE* is significantly negative (coefficient = −0.363, *t* statistic = −2.59), which indicates that the interaction between *Q* and *FRAG* decreases from 0.391 to 0.018 across the deciles of *RKINSTRADE*.

7. Analyses Addressing Alternative Explanations

7.1. Analysis of Real Effects of Market Fragmentation

One alternative explanation for a higher association between investment and investment opportunities when market fragmentation is high is that market fragmentation reduces noise in the measure of investment opportunities. That is, there is no change in firms' investment but the empirical association between investment and investment opportunities is higher because there is less measurement error in investment opportunities. If this were the sole explanation for the higher association between investment and investment opportunities when market fragmentation is high, we would not expect to observe real effects of market fragmentation. To examine this possibility, we test whether market fragmentation is associated with higher levels of future profitability, controlling for other factors. We do so by estimating the following regression model:

$$\begin{aligned} &ROA_{i,t+1} \text{ or } \Sigma^3{}_{i=1} ROA_{i,t+i} \\ &\quad = b_0 + b_1 FRAG_{it} + b_2 ROA_{it} + b_3 SIZE_{it} \\ &\qquad + \mathit{Firm\ Fixed\ Effects} + \mathit{Year\ Fixed\ Effects} + e_{i,t+i}, \end{aligned} \tag{11a}$$

where the dependent variable is net income before extraordinary items divided by total assets, measured either in the subsequent fiscal year or as the sum over the subsequent three fiscal years. If market fragmentation has real effects on investment, then we predict b_1 is positive.

We also examine whether market fragmentation is associated with the profitability of capital investment decisions by estimating the following model:

$$\begin{aligned} &ROA_{i,t+1} \text{ or } \Sigma^3{}_{i=1} ROA_{i,t+i} \\ &\quad = b_0 + b_1 FRAG_{it} + b_2 FRAG_{it} \times CAPX_{it} + b_3 CAPX_{it} \\ &\qquad + b_4 ROA_{it} + b_5 SIZE_{it} + \mathit{Firm\ Fixed\ Effects} \\ &\qquad + \mathit{Year\ Fixed\ Effects} + e_{i,t+i}, \end{aligned} \tag{11b}$$

where b_3 captures the return on capital expenditures when market fragmentation equals zero, and b_2 captures the effect of market fragmentation on this return. If market fragmentation leads to more profitable investment decisions, we predict b_2 is positive.

Table 7 presents findings from the estimation of Equations (11a) and (11b). The findings in column (1) reveal that, conditional on current performance, one-year-ahead performance increases with market fragmentation. In particular, the coefficient on *FRAG* is significantly positive (coefficient = 0.019, *t* statistic = 3.33). The coefficient on *FRAG* in column (2) is also positive (coefficient = 0.042, *t* statistic = 2.13), indicating a similar association between market fragmentation and profitability over a longer period. The findings in columns (3) and (4) reveal that the return on capital investment increases with market fragmentation. Specifically, the coefficient on the interaction between *FRAG* and *CAPX* is positive (coefficient = 0.001, *t* statistic = 2.02 in column (3) and coefficient = 0.003, *t* statistic = 1.69 in column (4)). The findings presented in Table 7 suggest that market fragmentation has a real

Table 7. Market Fragmentation and Future Performance

Variable	(1) Dependent variable $= ROA_{i,t+1}$	(2) Dependent variable $= \Sigma^3_{i=1}\ ROA_{i,t+i}$	(3) Dependent variable $= ROA_{i,t+1}$	(4) Dependent variable $= \Sigma^3_{i=1}\ ROA_{i,t+i}$
$FRAG_{it}$	0.019***	0.042**	0.016**	0.033
	(3.33)	(2.13)	(2.55)	(1.52)
$FRAG_{it} \times CAPX_{it}$			0.001**	0.003*
			(2.02)	(1.69)
$CAPX_{it}$			0.000	−0.000
			(1.19)	(−0.46)
ROA_{it}	0.143***	0.258***	0.142***	0.258***
	(3.54)	(2.75)	(3.54)	(2.75)
$SIZE_{it}$	−0.008***	−0.030***	−0.008***	−0.033***
	(−3.19)	(−4.54)	(−3.32)	(−4.60)
Firm and year fixed effects	Yes	Yes	Yes	Yes
No. of observations	101,096	83,154	101,096	83,154
Adjusted R^2	0.570	0.700	0.570	0.700

Notes. Table 7 reports pooled OLS regression results from the estimation of Equations (11a) and (11b). Results from Equation (11a) are reported in columns (1) and (2), and results from Equation (11b) are reported in columns (3) and (4). *ROA* is net income before extraordinary items divided by total assets, *FRAG* equals one minus the Herfindahl index of the concentration of trading across trading venues during the fiscal year, *CAPX* is capital expenditures as a percentage of total assets, and *SIZE* is the natural log of total assets. *t* statistics based on standard errors clustered by firm are shown in parentheses.

***, **, and *Significance at the 0.01, 0.05, and 0.10 levels, respectively.

effect on firms' profitability, rather than simply decreasing measurement error in Tobin's *q*.[22]

7.2. Analyses to Improve the Identification of Market Fragmentation

We use two empirical approaches to mitigate the possibility that factors causing firms to have more fragmented trading also are associated with the responsiveness of investment to investment opportunities. The first approach is a difference-in-differences analysis around a regulatory change, that is, the introduction of Reg NMS in 2007, which increased market fragmentation. The second approach uses the number of U.S. trading venues as an instrument for market fragmentation.

As noted in Section 2.2.1, Reg NMS was introduced to increase competition in equity markets, thus promoting more efficient trading services and more efficient pricing of stocks. As documented by Haslag and Ringgenberg (2023), Reg NMS led to an improvement in market quality on average, particularly for large firms.[23] If the increase in market quality following Reg NMS led to greater revelatory price efficiency, then we expect a greater sensitivity of investment to investment opportunities following Reg NMS.

We use a four-year window to conduct our analysis of Reg NMS, which includes two years prior to the implementation of Reg NMS in 2007 and two years after and excludes the year of implementation. First, we test whether the increase in market fragmentation associated with the implementation of Reg NMS results in an increase in the responsiveness of investment to investment opportunities by estimating the following regression:

$$\begin{aligned} CAPX_{it} &= b_0 + b_1 Q_{i,t-1} + b_2 Q_{i,t-1} \times POST_t + b_3 POST_t \\ &\quad + b_4 CF_{it} + b_5 RETURN3_{it} + b_6 INVASSETS_{i,t-1} \\ &\quad + Firm\ Fixed\ Effects + Year\ Fixed\ Effects + e_{it}, \qquad (12a) \end{aligned}$$

where *POST* is an indicator variable that equals one for the two years after 2007 and zero otherwise. We predict b_2 is positive if firms have greater responsiveness of investment to investment opportunities after the implementation of Reg NMS in 2007.

We also estimate alternative versions of Equation (12a) that shift the event year backward and forward by two years. These analyses permit us to assess whether our findings are attributable to Reg NMS rather than underlying trends that affect the sensitivity of investment to investment opportunities independent of Reg NMS. In these tests, we predict b_2 is zero.

Because the two-year period following Reg NMS coincides with the global financial crisis, we also estimate a version of Equation (12a) that uses a four-quarter event window—two quarters prior to the implementation of Reg NMS is August 2007 and two quarters after, excluding the quarter of implementation. This allows us to distinguish the effects of Reg NMS from the effects of the global financial crisis.

The findings in column (1) of Table 8, Panel A, reveal that the sensitivity of investment to investment

Table 8. Analyses Addressing Alternative Explanations

Panel A: Analysis around Reg NMS

Variable	(1) 2007 ± 2 years	(2) Placebo Test 2005 ± 2 years	(3) Placebo Test 2009 ± 2 years	(4) 2007 ± 2 quarters
	Dependent variable = $CAPX_{it}$			
$Q_{i,t-1}$	0.792***	0.589***	0.824***	0.079***
	(8.75)	(7.35)	(8.65)	(2.83)
$Q_{i,t-1} \times POST_t$	0.213**	0.064	−0.080	0.055***
	(2.31)	(0.76)	(−1.01)	(2.74)
$POST_t$	−1.185***	−0.777**	−0.333	−0.280*
	(−3.26)	(−2.07)	(−0.98)	(−1.92)
CF_{it}	0.031***	0.020***	0.026***	−0.008
	(4.22)	(2.63)	(3.90)	(−0.67)
$RETURN3_{it}$	−0.267***	−0.271***	−0.087	−0.075*
	(−2.69)	(−3.57)	(−1.41)	(−1.65)
$INVASSETS_{i,t-1}$	0.051***	0.034***	0.057***	0.001
	(3.88)	(2.98)	(4.75)	(0.05)
Firm and year fixed effects	Yes	Yes	Yes	No
Firm and year-quarter fixed effects	No	No	No	Yes
No. of observations	17,879	18,238	16,703	18,239
Adjusted R^2	0.686	0.754	0.719	0.518

Panel B: Difference-in-difference analysis around Reg. NMS

Variable	(1) 2007 ± 2 years Dependent variable = $CAPX_{it}$
$Q_{i,t-1}$	0.732***
	(6.60)
$Q_{i,t-1} \times POST_t$	0.009
	(0.08)
$Q_{i,t-1} \times POST_t \times LARGE_{i,t-1}$	0.542***
	(2.65)
$Q_{i,t-1} \times LARGE_{i,t-1}$	0.827***
	(4.40)
$POST_t \times LARGE_{i,t-1}$	−0.568
	(−1.27)
$POST_t$	−1.089**
	(−2.07)
$LARGE_{i,t-1}$	−4.195***
	(−6.00)
CF_{it}	0.035***
	(4.12)
$RETURN3_{it}$	−0.286**
	(−2.50)
$INVASSETS_{i,t-1}$	0.079***
	(4.17)
Firm and year fixed effects	Yes
No. of observations	17,879
Adjusted R^2	0.745

Panel C: Instrumental variable analysis using the number of trading venues

Variable	(1) Dependent variable $= FRAG_{t-1}$	(2) Dependent variable $= Q_{t-1} \times FRAG_{t-1}$	(3) Dependent variable = $CAPX_t$
$log(\#VENUES)_{t-1}$	0.253***	1.344***	
	(13.55)	(9.45)	
$Q_{i,t-1} \times log(\#VENUES)_{t-1}$	−0.003***	0.040***	
	(−12.93)	(7.55)	
$Q_{i,t-1}$	0.007***	0.074***	0.616***
	(7.73)	(5.04)	(6.13)
$Pred.\ (Q_{i,t-1} \times FRAG_{i,t-1})$			1.081**
			(2.16)

Table 8. (Continued)

Panel C: Instrumental variable analysis using the number of trading venues			
Variable	(1) Dependent variable $= FRAG_{t-1}$	(2) Dependent variable $= Q_{t-1} \times FRAG_{t-1}$	(3) Dependent variable $= CAPX_t$
$Pred.\ FRAG_{i,t-1}$			1.252
			(0.28)
CF_{it}	0.000***	0.003***	0.029***
	(3.25)	(7.84)	(10.44)
$RETURN3_{it}$	−0.004***	−0.020***	−0.249***
	(−11.23)	(−7.62)	(−7.07)
$INVASSETS_{i,t-1}$	−0.000***	−0.003***	0.018***
	(−12.09)	(−10.61)	(5.42)
Firm and year fixed effects	Yes	Yes	Yes
Kleibergen-Paap Wald F statistic			102.41***
No. of observations	109,020	109,020	109,020
Adjusted R^2	0.908	0.727	0.592

Notes. Panels A and B present results from the estimation of Equations (12a) and (12b). Panel A displays the overall effect of Reg NMS on investment sensitivities, and Panel B displays results from a difference-in-differences analysis that uses small firms as a control group. In Panel A, column 1, and Panel B, *POST* is an indicator that equals one for the two years after the implementation of Reg NMS in 2007. In column 2 (3), *POST* is an indicator variable that equals one for the two years after 2005 (2009). In column 4, *POST* is an indicator variable that equals one for the two quarters after the implementation of Reg NMS. Panel C presents results from a two-stage instrumental variable estimation of Equations (13a) through (13c). *#VENUES* is the number of trading venues existing at the end of the firm's fiscal year. See Appendix A for all other variable descriptions. In column 3 of Panel C, t statistics based on bootstrapped standard errors from 1,000 estimations are shown in parentheses. All other t statistics are based on standard errors clustered by firm.

***, **, and *Significance at the 0.01, 0.05, and 0.10 levels, respectively.

opportunities increases following the introduction of Reg NMS. In particular, the coefficient on Q increases by 0.213 (t statistic = 2.31), which represents a 26.9% increase from 0.792 prior to Reg NMS to 1.005 following Reg NMS.[24]

The findings in columns (2) and (3) relate to estimations of Equation (12a) in which we shift the sample and *POST* indicator variable forward or backward by two years. In both cases, we do not observe a significant difference in the sensitivity of investment to investment opportunities, that is, the coefficients on the interactions of Q and *POST* are not significantly different from zero.

The inferences based on the findings based on quarterly data in column (4) are similar to those based on the findings in column (1). The sensitivity of investment to investment opportunities at the quarterly level (Q coefficient = 0.079, t statistic = 2.83) significantly increases following Reg NMS ($Q \times POST$ coefficient = 0.055, t statistic = 2.74). Taken together, the findings in Panel A suggest that the increase in the sensitivity of investment to investment opportunities resulted from an event in 2007 (i.e., Reg NMS) rather than from a general time trend.

Our difference-in-differences test compares the effect of Reg NMS on large firms to that on small firms. Prior research documents that market fragmentation benefits primarily large firms (Degryse et al. 2015, Gresse 2017, Haslag and Ringgenberg 2023). Thus, we use small firms as a control group, and we calculate the effect of Reg NMS on large firms relative to small firms by estimating the following regression[25]:

$$\begin{aligned} CAPX_{it} &= b_0 + b_1 Q_{i,t-1} + b_2 Q_{i,t-1} \times POST_t \\ &\quad + b_3 Q_{i,t-1} \times POST_t \times LARGE_{i,t-1} \\ &\quad + b_4 Q_{i,t-1} \times LARGE_{i,t-1} + b_5 POST_t \times LARGE_{i,t-1} \\ &\quad + b_6 POST_t + b_7 LARGE_{i,t-1} + b_8 CF_{it} + b_9 RETURN3_{it} \\ &\quad + b_{10} INVASSETS_{i,t-1} + Firm\ Fixed\ Effects \\ &\quad + Year\ Fixed\ Effects + e_{it}, \end{aligned} \tag{12b}$$

where *LARGE* is a variable that indicates whether the firm's assets are above the sample median. We predict b_3 is positive if large firms have a greater change in the sensitivity of investment to investment opportunities than small firms after the implementation of Reg NMS in 2007.

The findings presented in Table 8, Panel B, reveal that the increase in sensitivity of investment to investment opportunities following the introduction of Reg NMS is greater for large firms than for small firms. In particular, the increase in the coefficient on Q is greater for large firms, that is, the $Q \times POST \times LARGE$ coefficient, 0.542, is significantly positive (t statistic = 2.65).

Next, we address endogeneity by removing firm characteristics that may simultaneously affect both market fragmentation and the sensitivity of investment to investment opportunities. We use the number of trading venues in the United States as an instrument for *FRAG*, following Haslag and Ringgenberg (2023).[26] We use a two-stage approach, where the first-stage equations, Equations (13a) and (13b), regress *FRAG* and $Q \times FRAG$ on the natural log of the number of trading venues that exist at year end and its interaction with *Q*, along with the control variables from Equation (13c) to help ensure that coefficient estimates from the estimation of Equation (13c) are consistent (Wooldridge 2002). The second-stage equation, Equation (13c), re-estimates Equation (2a) after replacing *FRAG* with the predicted values of *FRAG* and $Q \times FRAG$ from Equations (13a) and (13b):

$$\begin{aligned} \text{Stage 1}: \ & FRAG_{i,t-1} \\ &= a_0 + a_1 log(\#VENUES)_{t-1} \\ &\quad + a_2 Q_{i,t-1} \times log(\#VENUES)_{t-1} + a_3 Q_{i,t-1} \\ &\quad + a_4 CF_{it} + a_5 RETURN3_{it} + a_6 INVASSETS_{i,t-1} \\ &\quad + Firm\ Fixed\ Effects + Year\ Fixed\ Effects + e_{i,t-1}, \end{aligned} \tag{13a}$$

$$\begin{aligned} & Q_{i,t-1} \times FRAG_{i,t-1} \\ &= a_0 + a_1 log(\#VENUES)_{t-1} + a_2 Q_{i,t-1} \times log(\#VENUES)_{t-1} \\ &\quad + a_3 Q_{i,t-1} + a_4 CF_{it} + a_5 RETURN3_{it} + a_6 INVASSETS_{i,t-1} \\ &\quad + Firm\ Fixed\ Effects + Year\ Fixed\ Effects + e_{i,t-1}, \end{aligned} \tag{13b}$$

$$\begin{aligned} \text{Stage 2}: \ & CAPX_{it} \\ &= b_0 + b_1 Q_{i,t-1} + b_2 Pred.\ Q_{i,t-1} \times FRAG_{i,t-1} \\ &\quad + b_3 Pred.\ FRAG_{i,t-1} + b_4 CF_{it} + b_5 RETURN3_{it} \\ &\quad + b_6 INVASSETS_{i,t-1} + Firm\ Fixed\ Effects \\ &\quad + Year\ Fixed\ Effects + e_{it}. \end{aligned} \tag{13c}$$

The variable *log(#VENUES)* is the natural log of the number of trading venues existing in the United States at the end of the firm's fiscal year. In Equation (13a), we expect b_1 is positive if the level of fragmentation increases with the number of trading venues. Likewise, we expect b_2 is positive in Equation (13b). As with Equation (2a), we predict b_2 in Equation (13c) is positive if market fragmentation is associated with an increase in RPE that informs either managers or creditors about the firm's investment opportunities. Because *Pred. FRAG* is estimated with error, when estimating Equation (13c), we bootstrap standard errors based on the empirical distribution of coefficient estimates across 1,000 estimations of the model.

The findings in column (1) of Table 8, Panel C, relating to the estimation of Equation (13a), reveal that *FRAG* is increasing in the number of markets (coefficient = 0.253, *t* statistic = 13.55). The findings in column (2) reveal that the interaction of *Q* and *FRAG* is increasing in the interaction of *Q* and the number of markets (coefficient = 0.040, *t* statistic = 7.55). Finally, the results from the second-stage regression presented in column (3) reveal the same inferences as those based on the findings presented in Table 3. In particular, the coefficient on the interaction between *Q* and *Pred. FRAG* is significantly positive (coefficient = 1.081, *t* statistic = 2.16). The Kleibergen-Paap statistic rejects the null hypothesis that Equation (13c) is underidentified, providing support for the instruments we used in first-stage regressions.

Taken together, the findings presented in Table 8 increase our confidence in the conclusion that market fragmentation leads to improvement in price efficiency that facilitates corporate investment. However, we acknowledge that limitations inherent to these tests prevent us from conclusively drawing causal inferences.

8. Summary and Conclusions

This study examines how equity market fragmentation affects firms' capital investment decisions. Recent empirical research finds that market fragmentation improves market quality, as reflected by, for example, lower bid-ask spreads. We examine whether this increase in market quality translates into greater revelatory price efficiency. Market fragmentation can lead to an increase in the sensitivity of capital investment to investment opportunities by leading to prices that reveal with greater precision information to managers, that is, the managerial learning channel, or to creditors, that is, the financing channel, about their firms' investment opportunities.

Findings reveal that the association between capital investment and investment opportunities is increasing in market fragmentation. The sensitivity of investment to investment opportunities increases by 6.8% across the interquartile range of market fragmentation. In addition, findings reveal that effect of market fragmentation on the sensitivity of investment to investment opportunities is greater when investors are more heterogeneous.

These findings are consistent with market fragmentation leading to prices that reveal with greater precision information either to managers or to creditors about the firm's investment opportunities. Inferences based on an instrumental variable test and a difference-in-differences test are consistent with those based on our primary findings. To determine the extent to which one or both of the channels contributes to the increase in sensitivity of capital investment to investment opportunities, we compare the effect of market fragmentation on the sensitivity of investment to investment opportunities for firms with low and high levels of financing constraints. Findings reveal that the effects of market fragmentation on investment increase with financing constraints, thereby providing support for the financing channel. Additional findings indicate that the effect of market fragmentation on investment is weaker when firms finance investment primarily through private debt from creditors who likely have access to a richer set of information beyond stock prices, suggesting that the financing channel works through public debt. We find evidence that the effects of market fragmentation also work through the managerial learning channel, as the effects are evident when financing constraints are low and are larger in firms whose executives are less informed.

Our findings indicate one mechanism by which market fragmentation increases revelatory price efficiency. Because market fragmentation is associated with increased competition among trading venues and lower trading costs, investors have greater opportunities to benefit from informed trade. Consistent with this notion, we find that market fragmentation is associated with higher levels of information acquisition by investors, an increase in informed trading, and stock prices that more fully reflect future earnings news.

Taken together, our study's findings provide the first empirical evidence in the academic literature of real effects of market fragmentation on corporate decision making. Our results suggest that market fragmentation leads to an improvement in revelatory price efficiency, which informs managers and particularly lenders about a firm's investment opportunities.

Acknowledgments
The authors thank Xanthi Gkougkousi, the associate editor, two anonymous reviewers, Frank Zhou (discussant), Kurt Gee (discussant), and workshop participants at the 2022 Financial Accounting and Reporting Section Midyear Meeting, the 2022 Hawaii Accounting Research Conference, Louisiana State University, Southern Methodist University, the University of California, Berkeley, and the

Appendix A. Variable Definitions

Variable	Definition
FRAG	One minus the Herfindahl index of trade value across all trading venues that exist during the calendar year. FRAG ranges from 0 (no fragmentation) to 1 (high fragmentation).
CAPX	Capital expenditures as a percentage of beginning-of-year total assets.
Q	Tobin's q, equal to the sum of market value of equity and book value of debt, divided by the beginning-of-year book value of total assets (Tobin 1969).
CF	Net operating cash flow as a percentage of beginning-of-year total assets.
RETURN3	The value-weighted, market-adjusted stock return over the current and subsequent two fiscal years.
INVASSETS	Inverse of total assets.
SIZE	Natural log of total assets.
MB	Market value of equity divided by the book value of equity.
ROA	Income before extraordinary items, divided by year-end total assets.
LEVERAGE	Sum of long-term and short-term debt, divided by the sum of market value of equity and book value of debt.
IDIOVOL	The variance of the residual obtained by fitting the Carhart (1997) four-factor model to daily stock returns over the fiscal year.
TURNOVER	Natural log of trading volume scaled by the number of shares outstanding.
ANALYSTS	Natural log of one plus the number of analysts on I/B/E/S who issue an earnings forecast during the fiscal year.
INSTOWN	Percentage of common shares owned by institutional investors as reported in 13F filings.
NYSE	Indicator equal to 1 if the firm's primary listing venue is the New York Stock Exchange.
NASDAQ	Indicator equal to 1 if the firm's primary listing venue is the Nasdaq Stock Market.
EDGAR_10KQ	The natural log of the number of the firm's 10-K or 10-Q filings downloaded from EDGAR during the fiscal year, using the SEC's EDGAR log file.
EDGAR_ALL	The natural log of the total number of the firm's filings downloaded from EDGAR during the fiscal year, using the SEC's EDGAR log file.
PIN_EKO	Probability of informed trade, measured following Easley et al. (1996) and obtained from Stephen Brown (http://www.rhsmith.umd.edu/faculty/sbrown/).
PIN_OWR	Probability of informed trade, measured following Odders-White and Ready (2008) and obtained from Edwin Hu (https://edwinhu.github.io/).
INVPRICE	Inverse of stock price.
RETURN	The buy-and-hold stock return over the fiscal year.
ΔEARN	Change in annual net income before extraordinary items, scaled by the market value of equity.
RKINVESTHET	Decile rank of one minus a Herfindahl index of the proportion of holdings of various investor types defined in Bushee (1998): noninstitutional investors and transient, quasi-indexer, and dedicated institutional investors, scaled between 0 and 1.
RKCONSTR	Decile rank of financing constraints, scaled between 0 and 1. Financing constraints are measured using the Hoberg and Maksimovic (2015) "Delay Investment Score".
RKAQ	Decile rank of negative 1 times the standard deviation of firm-level residuals from the Dechow and Dichev (2002) model over years $t-5$ to $t-1$. The variable is scaled between 0 and 1.
RKPRIVATE	Decile rank of the ratio of private debt to the sum of private debt and corporate bonds, using data from Capital IQ and scaled between 0 and 1. Private debt is the sum of total revolving credit and total term loans, and corporate bonds is the sum of total senior bonds and notes and total subordinated bonds and notes.

Appendix A. (Continued)

Variable	Definition
DEBTAMT	Natural log of the total dollar amount of bonds issued during the fiscal year.
DEBTYIELD	Weighted average yield on bonds issued during the year.
RKINSTRADE	Decile rank of the profitability of insiders' trades, scaled between 0 and 1. The profitability of insiders' trades in year t is measured as the average one-month, Carhart (1997) four-factor risk-adjusted return following each insider's transaction.
#VENUES	Natural log of the number of trading venues existing in the United States at the end of the firm's fiscal year.

Appendix B. Trading Volume Across Exchanges

2017 trading volume	All stocks (N = 3,536)	NYSE stocks (N = 1,304)	Nasdaq stocks (N = 2,079)	Other stocks (N = 153)
NYSE	18.1%	30.0%	0.0%	0.0%
NYSE American	0.3%	0.1%	0.1%	16.0%
NYSE Arca	6.0%	5.6%	6.5%	9.7%
National Stock Exchange	0.0%	0.0%	0.0%	0.1%
Chicago Stock Exchange	0.4%	0.4%	0.3%	0.1%
Direct Edge A	1.5%	1.5%	1.4%	1.2%
Direct Edge X	6.0%	5.0%	7.5%	8.0%
BATS Y	4.4%	4.5%	4.4%	3.0%
BATS Z	5.2%	5.2%	5.2%	4.5%
NASDAQ	17.6%	9.3%	30.9%	8.1%
NASDAQ OMX BX	3.2%	3.1%	3.3%	1.9%
NASDAQ OMX PSX	0.6%	0.5%	0.9%	0.3%
Investors Exchange	2.3%	2.4%	2.2%	1.9%
Off-exchange (FINRA)	34.5%	32.4%	37.3%	45.4%

Notes. Appendix B presents the distribution of trades of common shares across trading venues for the calendar year 2017. The 14 trading venues include 13 exchanges and an aggregation of trades from various nonexchange trading venues (e.g., dark pools), which we obtain from FINRA. "NYSE Stocks" ("Nasdaq Stocks") includes only stocks that initially listed on the New York Stock Exchange (Nasdaq Stock Exchange). Stocks that initially listed on another trading venue are included in "Other Stocks."

University of Technology Sydney for providing helpful comments and suggestions.

Endnotes

[1] For example, whereas 30 years ago U.S. equity trading occurred primarily on the New York Stock Exchange (NYSE), the NYSE currently accounts for less than 20% of the trading volume of U.S. equities (CBOE 2021).

[2] Historically, the NYSE was the only U.S. exchange that did not allow trading of stocks listed on rival U.S. exchanges. The NYSE ended that restriction in April 2018 (Osipovich 2018).

[3] The Markets in Financial Instruments Directive similarly encouraged competition among European markets by allowing new trading venues to compete with traditional stock exchanges.

[4] For more detail, see https://www.sec.gov/rules/final/34-51808.pdf.

[5] O'Hara and Ye (2011) explains that market fragmentation could also harm market quality by reducing the liquidity available not only in individual markets but also in aggregate. This could occur if market fragmentation reduces the enforcement of time priority across markets, thereby disincentivizing traders from posting limit orders, or if access to trading venues is limited for some traders.

[6] It is likely that the sensitivity to trading costs increases with the frequency of trade, which implies the sensitivity to trading costs is greater for short-term investors. However, the literature provides evidence that both short-term and long-term investors trade on fundamental information that may be relevant to corporate investment decisions (Porter 1992, Lang and McNichols 1997, Ke and Petroni 2004).

[7] We consider debt financing but not equity financing for two reasons. First, we are interested in the transfer of information across, rather than within groups, that is, from stock prices, which are set by equity investors, to managers and creditors. Second, publicly traded firms typically obtain more debt than equity financing and do so more regularly. To the extent market fragmentation reduces the cost of equity, we would expect a substitution from debt-to-equity financing, which would bias against our finding evidence of stock prices informing creditors.

[8] It is possible that market fragmentation leads to information acquisition, which in turn leads to information asymmetry among investors. In this case, a higher equity cost of capital could constrain investment, leading to a lower sensitivity of investment to investment opportunities. This effect of market fragmentation, to the extent it occurs, is likely to bias against our hypothesis.

[9] We calculate market fragmentation across the 14 trading venues, following prior research (Haslag and Ringgenberg 2023). Inferences are unchanged when we calculate market fragmentation across only the 13 exchanges or when controlling for the extent of off-exchange trading. The former result is not surprising given the high correlation between the two measures of market fragmentation (0.93).

[10] We estimate variations of Equation (2a) using two alternative measures of investment—the sum of capital expenditures and research and development expenditures and the change in total assets. Untabulated analyses reveal that our primary inferences are unchanged when using these measures.

[11] For ease of exposition, we use the same notation for coefficients across equations. In all likelihood, they differ.

[12] Throughout the study, we use a 0.05 significance level under a two-sided alternative.

[13] Because prior research (Degryse et al. 2015, Gresse 2017, Haslag and Ringgenberg 2023) finds that improvements in market quality resulting from increased market fragmentation are concentrated primarily in large firms, we estimate a version of Equation (2a) that allows the effect of market fragmentation to vary with firm size. Untabulated findings reveal that the effects of market fragmentation on the sensitivity of investment to investment opportunities and to cash flows increase with firm size. In particular, the coefficient on the three-way interaction among *Q*, *FRAG*, and the decile rank of total assets is significantly positive (coefficient = 0.871, *t* statistic = 3.34).

[14] To test whether the association between market fragmentation and the sensitivity of investment to investment opportunities is evident in the cross section, and not simply a result of increasing market fragmentation over time, we estimate year-by-year regressions of Equation (2a). Consistent with our primary findings, the coefficient on the interaction between *Q* and *FRAG* is significantly positive (mean coefficient = 0.442, Fama-MacBeth (Fama and MacBeth 1973) *t* statistic = 2.68). Further, because the associations we observe in Table 3 could result from correlated omitted variables, we estimate a version of Equation (2a) that includes controls for several firm characteristics that may affect both market fragmentation and the sensitivity of investment to investment opportunities. In particular, we include the variables from Equation (1), both as main effects and interactions with the $Q_{t-1} \times FRAG_t$ term. Consistent with the results in Table 3, Panel A, findings presented in Table A1 of the online appendix reveal that the sensitivity of investment to investment opportunities increases significantly with market fragmentation in the presence of linear (*t* statistic = 2.20) or both linear and interactive controls (*t* statistic = 2.28).

[15] Although information contained in EDGAR downloads is presumably already known to managers, investors with superior information processing ability may generate new, value-relevant information from public filings (Engelberg et al. 2012).

[16] We thank Stephen Brown for providing our measure of *PIN_EKO* (available at http://www.rhsmith.umd.edu/faculty/sbrown/), and we thank Edwin Hu for providing our measure of *PIN_OWR* (available at https://edwinhu.github.io/pin/).

[17] Hoberg and Maksimovic (2015) constructs the measure using a textual analysis of the MD&A section of firms' 10-K filings. Firms with a higher delay investment score are more likely to be at risk for delaying their investments because of liquidity concerns.

[18] Inferences are unchanged when we estimate Equation (5) using an alternative measure of financing constraints based on the Kaplan and Zingales (1997) index of financial constraints, updated following Hadlock and Pierce (2010). Untabulated findings reveal that the coefficient on the three-way interaction among *Q*, *FRAG*, and *RKCONSTR* is 0.574 (*t* statistic = 2.86).

[19] Although this finding is consistent with market fragmentation working through the financing channel, it cannot rule out the managerial learning channel because improved financial disclosures may also facilitate managers' learning from stock prices (Jayaraman and Wu 2019).

[20] The negative coefficients on *FRAG* in column (1) of Panel D and in column (2) of Table 3 indicate lower levels of borrowing and investment when market fragmentation is low. Our focus is not the main effect of *FRAG* because a firm could have a higher value for *FRAG* but choose not to borrow and invest if profitable opportunities do not exist. Our predictions relate to the ability of a firm to borrow and invest when opportunities arise rather than to the average level of borrowings or investment.

[21] We estimated Equations (9a) and (9b) on a sample of private debt issues. Untabulated findings reveal a significantly positive but smaller coefficient on the interaction coefficient, b_2, in Equation (9a), and, contrary to expectations, a positive coefficient on b_1 in Equation (9b). These findings are consistent with the effects of market fragmentation working primarily through the public debt financing channel.

[22] Inferences are unchanged when we replace profitability in Equations (11a) and (11b) with total factor productivity, calculated following Bennett et al. (2020). In Equation (11a), the coefficient on total factor productivity is significantly positive for both one-year-ahead (*t* statistic = 1.95) and three-year-ahead (*t* statistic = 2.45) horizons. In Equation (11b), the coefficient on the interaction between market fragmentation and total factor productivity is significantly positive for both one-year-ahead (*t* statistic = 2.29) and three-year-ahead (*t* statistic = 2.46) horizons.

[23] It is also possible that the introduction of Reg NMS, by lowering trading costs for retail investors, led to an increase in noise trading and therefore a decrease in market quality and revelatory price efficiency. However, the findings in Haslag and Ringgenberg (2023) indicate an increase in market quality, on average.

[24] We also estimate versions of Equations (3)–(5) using Reg NMS as an instrument for market fragmentation. Untabulated findings reveal that coefficients on the instrument are significantly positive for Equations (3) and (5) but not significantly different from zero for Equation (4).

[25] O'Hara and Ye (2011) finds mixed evidence regarding the net benefits of market fragmentation for large versus small stocks. In particular, the study finds that market fragmentation benefits large NYSE-listed stocks and small Nasdaq-listed stocks. To the extent that the benefits of market fragmentation accrue to small firms, it will be harder to detect the effect of market fragmentation in this difference-in-differences test.

[26] A valid instrument needs to satisfy two conditions: the relevance condition and the exclusion condition. Regarding the relevance condition, we expect the level of market fragmentation to increase with the number of available trading venues. The Kleibergen-Paap Wald test rejects the null hypothesis of a weak instrument (*F* statistic = 102.41). The exclusion condition requires that the number of markets available for equity trading is related to firm-level capital investment only through fragmentation. Following Haslag and Ringgenberg (2023), we do not expect market centers to open or close based on the investment decisions of individual firms.

References

Aitken M, Chen H, Foley S (2017) The impact of fragmentation, exchange fees and liquidity provision on market quality. *J. Empirical Finance* 41:140–160.

Baker M, Wurgler J (2002) Market timing and capital structure. *J. Finance* 57:1–32.

Bakke TE, Whited TM (2010) Which firms follow the market? An analysis of corporate investment decisions. *Rev. Financial Stud.* 23(5):1941–1980.

Bennett B, Stulz R, Wang Z (2020) Does the stock market make firms more productive? *J. Financial Econom.* 136:281–306.

Biddle GC, Hilary G (2006) Accounting quality and firm-level capital investment. *Accounting Rev.* 81(5):963–982.

Biddle GC, Hilary G, Verdi RS (2009) How does financial reporting quality relate to investment efficiency? *J. Accounting Econom.* 48(2–3):112–131.

Blankespoor E, Linsmeier TJ, Petroni KR, Shakespeare C (2013) Fair value accounting for financial instruments: Does it improve the association between bank leverage and credit risk? *Accounting Rev.* 88(4):1143–1177.

Bo H, Lensink R, Sterken E (2003) Uncertainty and financing constraints. *Rev. Finance* 7(2):297–321.

Bond P, Edmans A, Goldstein I (2012) The real effects of financial markets. *Annu. Rev. Financial Econom.* 4(1):339–360.

Bond SR, Klemm A, Newton-Smith R, Syed M, Vlieghe GW (2004) The roles of expected profitability, Tobin's Q and cash flow in econometric models of company investment. Working paper, Nuffield College, Institute for Fiscal Studies, International Monetary Fund, Bank of England.

Bushee BJ (1998) The influence of institutional investors on myopic R&D investment behavior. *Accounting Rev.* 73(3):305–333.

Carhart MM (1997) On persistence in mutual fund performance. *J. Finance* 52(1):57–82.

CBOE (2021) U.S. Equities Market Volume Summary. Accessed September 6, 2021, https://www.cboe.com/us/equities/market_share/.

Chen D, Duffie D (2021) Market fragmentation. *Amer. Econom. Rev.* 111(7):2247–2274.

Chen H, Cohen L, Gurun U, Lou D, Malloy C (2020) IQ from IP: Simplifying search in portfolio choice. *J. Financial Econom.* 138(1): 118–137.

Chen Q, Goldstein I, Jiang W (2007) Price informativeness and investment sensitivity to stock price. *Rev. Financial Stud.* 20(3): 619–650.

Chowdhry B, Nanda V (1991) Multimarket trading and market liquidity. *Rev. Financial Stud.* 4(3):483–511.

Chung KH, Chuwonganant C (2012) Regulation NMS and market quality. *Financial Management* 41(2):285–317.

Crane A, Crotty K, Umar T (2023) Hedge funds and public information acquisition. *Management Sci.* 69(6):3241–3262.

Damodoran A, Liu CH (1993) Insider trading as a signal of private information. *Rev. Financial Stud.* 6:79–119.

Dávila E, Parlatore C (2021) Trading costs and informational efficiency. *J. Finance* 76(3):1471–1539.

De Fontnouvelle P, Fishe RP, Harris JH (2003) The behavior of bid-ask spreads and volume in options markets during the competition for listings in 1999. *J. Finance* 58(6):2437–2463.

Dechow PM, Dichev ID (2002) The quality of accruals and earnings: The role of accrual estimation errors. *Accounting Rev.* 77(s-1):35–59.

Degryse H, De Jong F, Kervel VV (2015) The impact of dark trading and visible fragmentation on market quality. *Rev. Finance* 19(4): 1587–1622.

deHaan E 2021. Using and interpreting fixed effects models. Working paper, Stanford Graduate School of Business, Stanford.

Downing C, Underwood S, Xing Y (2009) An intraday analysis of the relative informational efficiency of stocks and bonds. *J. Financial Quant. Anal.* 44(5):1081–1102.

Drake M, Roulstone D, Thornock J (2015) The determinants and consequences of information acquisition via EDGAR. *Contemporary Accounting Res.* 32:1128–1161.

Easley D, Kiefer NM, O'Hara M, Paperman JB (1996) Liquidity, information, and infrequently traded stocks. *J. Finance* 51(4): 1405–1436.

Engelberg JE, Reed AV, Ringgenberg MC (2012) How are shorts informed? Short sellers, news, and information processing. *J. Financial Econom.* 105(2):260–278.

Ettredge M, Kwon S, Smith D, Zarowin P (2005) The impact of SFAS No. 131 business segment data on the market's ability to anticipate future earnings. *Accounting Rev.* 80:773–804.

Fama EF, MacBeth JD (1973) Risk, return, and equilibrium: Empirical tests. *J. Political Econom.* 81(3):607–636.

Fazzari S, Hubbard RG, Petersen B (1988) Investment, financing decisions, and tax policy. *Amer. Econom. Rev.* 78(2):200–205.

Félez-Vinas E (2017) Market fragmentation, mini flash crashes and liquidity. Working paper, Stockholm Business School, Stockholm.

Ferracuti E, Stubben S (2019) The role of financial reporting in resolving uncertainty about corporate investment opportunities. *J. Accounting Econom.* 68(2–3). https://www.sciencedirect.com/journal/journal-of-accounting-and-economics/vol/68/issue/2.

Foucault T, Fresard L (2014) Learning from peers' stock prices and corporate investment. *J. Financial Econom.* 111(3):554–577.

Foucault T, Menkveld AJ (2008) Competition for order flow and smart order routing systems. *J. Finance* 63(1):119–158.

Francis J, LaFond R, Olsson P, Schipper K (2005) The market pricing of accruals quality. *J. Accounting Econom.* 39(2):295–327.

Gebhardt W, Hvidkjaer S, Swaminathan B (2005) Stock and bond market interaction: Does momentum spill over? *J. Financial Econom.* 75(3):651–690.

Gibbons B, Iliev P, Kalodimos P (2020) Analyst information acquisition via EDGAR. *Management Sci.* 67(2):769–793.

Goldstein I, Liu B, Yang L (2021) Market feedback: Evidence from the horse's mouth. Working paper, University of Pennsylvania, National Bureau of Economic Research, Tsinghua University, and University of Toronto.

Goldstein I, Yang S, Zuo L (2020) *The Real Effects of Modern Information Technologies (No. w27529)* (National Bureau of Economic Research, Cambridge, MA).

Gresse C (2017) Effects of lit and dark market fragmentation on liquidity. *J. Financial Marketing* 35:1–20.

Hadlock CJ, Pierce JR (2010) New evidence on measuring financial constraints: Moving beyond the KZ index. *Rev. Financial Stud.* 23(5):1909–1940.

Harris M, Raviv A (1993) Differences in opinion make a horse race. *Rev. Financial Stud.* 6:473–506.

Haslag PH, Ringgenberg M (2023) The demise of the NYSE and NASDAQ: Market quality in the age of market fragmentation. *J. Financial Quant. Anal.* Forthcoming.

Hayek FA (1945) The use of knowledge in society. *Amer. Econom. Rev.* 35(4):519–530.

Hoberg G, Maksimovic V (2015) Redefining financial constraints: A text-based analysis. *Rev. Financial Stud.* 28(5):1312–1352.

Jagolinzer AD, Larcker DF, Taylor DJ (2011) Corporate governance and the information content of insider trades. *J. Accounting Res.* 49(5):1249–1274.

Jayaraman S, Wu JS (2019) Is silence golden? Real effects of mandatory disclosure. *Rev. Financial Stud.* 32(6):2225–2259.

Kaplan SN, Zingales L (1997) Do investment-cash flow sensitivities provide useful measures of financing constraints? *Quart. J. Econom.* 112(1):169–215.

Ke B, Petroni K (2004) How informed are actively trading institutional investors? Evidence from their trading behavior before a break in a string of consecutive earnings increases. *J. Accounting Res.* 42(5):895–927.

Ke B, Huddart S, Petroni K (2002) What insiders know about future earnings and how they use it: Evidence from insider trades. *J. Accounting Econom.* 35:315–346.

Lang M, McNichols M (1997) *Institutional Trading and Corporate Performance (No. 1460)* (Graduate School of Business, Stanford University, Stanford, CA).

Lee C, Ma P, Wang C (2015) Search-based peer firms: Aggregating investor perceptions through Internet co-searches. *J. Financial Econom.* 116:410–431.

Loughran T, McDonald B (2017) The use of EDGAR filings by investors. *J. Behav. Finance* 18:231–248.

Loughran T, Ritter J (1995) The new issues puzzle. *J. Finance* 50:23–51.

Lundholm R, Myers L (2002) Bringing the future forward: The effect of disclosure on the returns-earnings relation. *J. Accounting Econom.* 40:809–839.

Malamud S, Rostek M (2017) Decentralized exchange. *Amer. Econom. Rev.* 107(11):3320–3362.

Mendelson H (1987) Consolidation, fragmentation, and market performance. *J. Financial Quant. Anal.* 22(2):189–207.

Meulbroek LK (1992) An empirical analysis of illegal insider trading. *J. Finance* 47:1661–1699.

Myers SC, Majluf NS (1984) Corporate financing and investment decisions when firms have information that investors do not have. *J. Financial Econom.* 13(2):187–221.

O'Hara M, Ye M (2011) Is market fragmentation harming market quality? *J. Financial Econom.* 100(3):459–474.

Odders-White ER, Ready MJ (2008) The probability and magnitude of information events. *J. Financial Econom.* 87(1):227–248.

Osipovich A (2018) NYSE opens doors to stocks from rival exchanges, ending decades-old policy. *Wall Street Journal* (April 9). https://www.wsj.com/articles/nyse-opens-doors-to-stocks-from-rival-exchanges-ending-decades-old-policy-1523281649.

Pagano M (1989) Trading volume and asset liquidity. *Quart. J. Econom.* 104:255–274.

Piotroski J, Roulstone D (2005) Do insider trades reflect both contrarian beliefs and superior knowledge about future cash-flow realizations? *J. Accounting Econom.* 39:55–81.

Porter ME (1992) Capital choices: Changing the way America invests in industry. *J. Appl. Corporate Finance* 5(2):4–16.

Redfearn B (2019) The nexus of equity market structure and investor relations: Remarks at the 2019 Annual Conference of the National Investor Relations Institute. Accessed November 11, 2019, https://www.sec.gov/news/speech/speech-redfearn-060319.

Seyhun HN (1992) Why does aggregate insider trading predict future stock returns? *Quart. J. Econom.* 107:1303–1331.

Seyhun HN (1998) *Investment Intelligence from Insider Trading* (MIT Press, Cambridge, MA).

Stoll HR (2001) Market fragmentation. *Financial Anal. J.* 57:16–20.

Tobin J (1969) A general equilibrium approach to monetary theory. *J. Money Credit Bank.* 1(1):15–29.

U.S. Securities and Exchange Commission (2005) *Regulation National Market System* (U.S. Securities and Exchange Commission, Washington, DC).

U.S. Securities and Exchange Commission (1975) *Securities Acts Amendments* (U.S. Securities and Exchange Commission, Washington, DC).

Wooldridge J (2002) *Econometric Analysis of Cross Section and Panel Data* (MIT Press, Cambridge, MA).

Ye M, Zheng M, Zhu W (2019) Price discreteness and investment to price sensitivity. Working paper, University of Illinois at Urbana-Champaign.